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PORTABLE PLAYBACK DEVICE POWER MANAGEMENT

Abstract

An example implementation of portable playback device power management involves launching a power coordinator background process, the power coordinator background process having multiple client programs and establishing respective inter-process communication (IPC) mechanisms between the multiple client programs and the power coordinator background process. The implementation further involves receiving, via the established IPC mechanisms from the multiple client programs, messages indicating that the respective client program is ready to suspend, and determining that each client program of the multiple client programs is ready to suspend. The implementation further includes sending instructions to the operating system to kernel suspend. While in kernel suspend, the playback device detects a particular trigger to kernel resume and in response, performs a kernel resume.

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Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS [0001] This application is a continuation of and claims priority to U.S. patent application Ser. No. 18,473,406 filed on Sep. 25, 2023 and published as U.S. Patent Application Publication No. 2024/0012463, which is a continuation of and claims priority to U.S. patent application Ser. No. 18/058,958, filed on Nov. 28, 2022 and now U.S. Pat. No. 11,809,257, which is a continuation of and claims priority to U.S. patent application Ser. No. 17/443,885, filed on Jul. 28, 2021 and now U.S. Pat. No. 11,513,580, which is a continuation of and claims priority to U.S. application Ser. No. 16/435,214, filed on Jun. 7, 2019 and now U.S. Pat. No. 11,093,016. This application is also related to U.S. patent application Ser. No. 16/435,235 now U.S. Pat. No. 11,126,243. The disclosures of each of the foregoing applications are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The present technology relates to consumer goods and, more particularly, to methods, systems, products, features, services, and other elements directed to voice-assisted control of media playback systems or some aspect thereof.

BACKGROUND

[0003] Options for accessing and listening to digital audio in an out-loud setting were limited until in 2002, when SONOS, Inc. began development of a new type of playback system. Sonos then filed one of its first patent applications in 2003, entitled “Method for Synchronizing Audio Playback between Multiple Networked Devices,” and began offering its first media playback systems for sale in 2005. The Sonos Wireless Home Sound System enables people to experience music from many sources via one or more networked playback devices. Through a software control application installed on a controller (e.g., smartphone, tablet, computer, voice input device), one can play what she wants in any room having a networked playback device. Media content (e.g., songs, podcasts, video sound) can be streamed to playback devices such that each room with a playback device can play back corresponding different media content. In addition, rooms can be grouped together for synchronous playback of the same media content, and/or the same media content can be heard in all rooms synchronously.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Features, aspects, and advantages of the presently disclosed technology may be better understood with regard to the following description, appended claims, and accompanying drawings where:

[0005] Features, aspects, and advantages of the presently disclosed technology may be better

understood with regard to the following description, appended claims, and accompanying drawings, as listed below. A person skilled in the relevant art will understand that the features shown in the drawings are for purposes of illustrations, and variations, including different and/or additional features and arrangements thereof, are possible.

[0006] FIG. 1A is a partial cutaway view of an environment having a media playback system configured in accordance with aspects of the disclosed technology.

[0007] FIG. 1B is a schematic diagram of the media playback system of FIG. 1A and one or more networks.

[0008] FIG. 1C is a block diagram of a playback device.

[0009] FIG. 1D is a block diagram of a playback device.

[0010] FIG. 1E is a block diagram of a network microphone device.

[0011] FIG. 1F is a block diagram of a network microphone device.

[0012] FIG. 1G is a block diagram of a playback device.

[0013] FIG. 1H is a partially schematic diagram of a control device.

[0014] FIGS. 1-I, 1J, 1K, and 1L are schematic diagrams of corresponding media playback system zones.

[0015] FIG. 1M is a schematic diagram of media playback system areas.

[0016] FIG. 2A is a front isometric view of a playback device configured in accordance with aspects of the disclosed technology.

[0017] FIG. 2B is a front isometric view of the playback device of FIG. 3A without a grille.

[0018] FIG. 2C is an exploded view of the playback device of FIG. 2A.

[0019] FIG. 3A is a block diagram of a portable playback device configured in accordance with aspects of the disclosed technology.

[0020] FIG. 3B is a front isometric view of a portable playback device.

[0021] FIG. 3C is a rear isometric view of the portable playback device of FIG. 3B.

[0022] FIG. 3D is a top view of the portable playback device of FIG. 3B.

[0023] FIG. 3E is a bottom view of the portable playback device of FIG. 3B.

[0024] FIG. 3F is an isometric view of a charging base configured to facilitate charging of the portable playback device of FIG. 3B.

[0025] FIGS. 4A, 4B, 4C, and 4D are schematic diagrams of a control device in various stages of operation in accordance with aspects of the disclosed technology.

[0026] FIG. 5 is front view of a control device.

[0027] FIG. 6 is a message flow diagram of a media playback system.

[0028] FIG. 7A is a functional block diagram illustrating an example power coordinator system in accordance with aspects of the disclosed technology.

[0029] FIG. 7B is a functional block diagram illustrating an example inter-process communication in accordance with aspects of the disclosed technology.

[0030] FIG. 7C is a timing diagram illustrating an example suspend algorithm in accordance with aspects of the disclosed technology.

[0031] FIG. 7D is a message flow diagram further illustrating the example suspend algorithm of FIG. 7C.

[0032] FIGS. 7E and 7F are timing diagrams illustrating exemplary kernel suspend and resume timing in accordance with aspects of the disclosed technology.

[0033] FIG. 7G is a table illustrating example power levels in accordance with aspects of the disclosed technology.

[0034] FIG. 8A is a functional block diagram illustrating example system architecture to facilitate wake triggers in accordance with aspects of the disclosed technology.

[0035] FIG. 8B is a block diagram showing an example hierarchy of components within an example portable playback device.

[0036] FIGS. 9A, 9B, 9C, 9D, 9E, and 9F are example flow diagrams illustrating example wake

trigger detection in accordance with aspects of the disclosed technology.

[0037] FIG. **10** is a table illustrating example power modes in accordance with aspects of the disclosed technology.

[0038] FIG. **11** is a flow diagram of an example method to facilitate power coordination in a portable playback device; and

[0039] FIG. **12** is a flow diagram of an example method to facilitate power management in a portable playback device.

[0040] The drawings are for purposes of illustrating example embodiments, but it should be understood that the inventions are not limited to the arrangements and instrumentality shown in the drawings. In the drawings, identical reference numbers identify at least generally similar elements. To facilitate the discussion of any particular element, the most significant digit or digits of any reference number refers to the Figure in which that element is first introduced. For example, element **103a** is first introduced and discussed with reference to FIG. **1A**.

DETAILED DESCRIPTION

I. Overview

[0041] Example techniques described herein involve power management techniques for portable playback devices. Example portable playback devices described herein include one or more speakers and audio amplifiers powerful enough to facilitate out loud audio playback at suitable volume levels in both exterior and interior environments. To maintain portability, these example playback devices include one or more batteries to power the portable playback device during audio playback when disconnected from AC power.

[0042] A challenge arising with such portable playback devices is maintaining a balance between audio output capabilities, battery life, and portability. Powering large enough speaker drivers and associated amplifiers to facilitate out loud audio playback at suitable volumes in exterior or large environments requires large capacity batteries or portable battery life will be undesirably brief. At the same time, increasing battery life by adding battery capacity adds undesirable weight and bulk to the portable playback devices.

[0043] In contrast to portable general purpose computing devices such as smartphones, tablets, and laptop computers, a portable playback device may support relatively few features. Example portable playback devices may support only audio playback and related functions (e.g., voice control) rather than be generally programmable with user-installed software (e.g., apps on a smartphone or tablet). Given the predictable range of functionality, the portable playback device may coordinate between programs responsible for various functions (e.g., audio playback, network connectivity, software upgrades) of the portable playback device to manage power use by the portable playback device, especially while disconnected from AC power.

[0044] Example portable playback devices may implement a power coordinator that coordinates between one or more client programs and the operating system (kernel) to determine when to suspend. Example suspend modes (referred to herein as kernel suspend) involve disabling the main processor(s) of the portable playback device, which suspends the kernel and userspace programs. In this mode, power use is drastically reduced as compared with normal (playback) mode. At the same time, time to resume playback from kernel suspend is drastically less than when the portable playback device is powered off, as the kernel and userspace programs do not need to be reloaded into memory during boot.

[0045] In an example, the power coordinator implemented as a background process and launched by the kernel during boot. After launching, the power coordinator establishes respective inter-process communication (IPC) mechanisms between multiple client programs and the power coordinator. These IPC mechanisms allow the client programs (responsible for various functionality of the portable playback device) to communicate whether they are ready to suspend and if so, the time at which they need to resume.

[0046] Accordingly, in operation, the power coordinator receives, via the established IPC

mechanisms from the multiple client programs, messages indicating that the respective client program is ready to suspend. Such messages may also indicate respective resume times for the various clients programs. Via these messages, the power coordinator monitors the respective ready (or not ready) to suspend states of the multiple client programs. When each client program of the multiple client programs is ready to suspend, the power coordinator sends instructions to the operating system to kernel suspend. The power coordinator may also set a kernel suspend timeout trigger to the earliest resume time among the resume times indicated in the received messages from the multiple client program so that the kernel is resumed at an appropriate time for all client programs.

[0047] While the power coordinator may coordinate suspend from userspace, the power coordinator is unable to resume the portable playback device from suspend, as userspace programs like the power coordinator and the client programs not executing during kernel suspend. Moreover, support for resuming the portable playback device on any of a number of different triggers, such as wake-on-button (e.g., a power button, to facilitate the user requesting a resume), wake-on-battery (e.g., based on a fault or change in status), wake-on-Bluetooth, wake-on-wireless, and/or wake-on-touch, may be desirable. To facilitate these multiple resume triggers, example portable playback devices may implement one or more secondary controllers (e.g., microcontrollers and/or system-on-chip) that monitor for the appropriate conditions corresponding to each resume trigger and cause the main processor(s) and kernel to resume in response to these resume triggers.

[0048] For instance, an example playback device may implement a power manager in firmware on a power management microcontroller. The power manager may monitor the battery for various conditions relating to wake-on-battery triggers, such as a battery fault (e.g., overheating), critical battery level, and/or battery charging status changed. Upon detecting one of these wake-on-battery triggers, the power manager causes the power management microcontroller to send an interrupt to wake up the main processor(s) and the kernel.

[0049] In another example, the example playback device may implement a Bluetooth Low Energy (BLE) kernel driver in a secondary SoC, which stays active during kernel resume. The BLE driver monitors the BLE interface for a successful BLE connection, which corresponds to a wake-on-BLE trigger. Upon detecting the wake-on-BLE trigger, the BLE kernel driver causes the secondary SoC to send an interrupt to wake up the main processor(s) and the kernel.

[0050] Since the power coordinator is not executing when the interrupt is received, the power coordinator is unable to determine the source of the kernel resume trigger. In example implementations, the kernel implements a power event queue to communicate with the power coordinator. During a kernel resume based on a given kernel resume trigger, the kernel adds a kernel resume event to the power event indicating the source of the kernel resume trigger.

[0051] When the power coordinator resumes, the power coordinator reads the power event queue to determine the source of the kernel resume trigger. The power coordinator shares this information with the client programs, which allows the client programs to take action or change behavior based on the source of the kernel resume trigger. For instance, if the source of the kernel resume trigger is wake-on-BLE, a client program responsible for audio playback may change the audio content source of the portable playback device to a BLE audio stream from the successful BLE connection that triggered the wake-on-BLE. Other examples are possible as well.

[0052] As described above, the portable playback device, via the power coordinator, gains consensus from the multiple client programs that they are each ready to suspend before suspending. At the same time, the portable playback device resumes quickly from a variety of resume triggers. In other words, the portable playback device is slow to suspend, but quick to wake up.

[0053] The power coordinator may use a similar technique to manage power level of the portable playback device. A “power level” refers to a specific configuration of settings that when set, results in a given hardware performance for a particular power cost. Example portable playback devices may support multiple power levels and transition between power levels in an attempt to match

current functions of the portable playback device to the most efficient power level.

[0054] In an example, in operation the multiple client programs determine their respective minimum power level. The client programs send messages indicating their respective power level requirements to the power coordinator via the established IPC mechanisms. The power coordinator monitors these messages, and determines a particular power level from among the multiple power levels, the particular power level being the highest power level among the respective power level requirements of the multiple client programs. The power coordinator then sends instructions to the operating system to operate at the particular power level. When the power coordinator detects that at least one client program requires a higher power level, the power coordinator instructs the kernel to increase the power level to support the requirements of this client program.

[0055] Example techniques related to portable playback device power management. An example implementation includes: launching a power coordinator background process, the power coordinator background process having multiple client programs; establishing respective inter-process communication (IPC) mechanisms between the multiple client programs and the power coordinator background process; receiving, via the established IPC mechanisms from the multiple client programs, messages indicating (a) that the respective client program is ready to suspend and (b) a respective resume time; determining, based on the received messages from the multiple client programs, that each client program of the multiple client programs is ready to suspend; based on determining that each client program of the multiple client programs is ready to suspend, (i) sending instructions to the operating system to kernel suspend and (ii) setting a kernel suspend timeout trigger to the earliest resume time among the resume times indicated in the received messages from the multiple client programs; while in the kernel suspend, detecting a particular trigger to kernel resume from among a plurality of triggers to kernel resume, wherein the plurality of triggers to kernel resume comprise the kernel suspend timeout trigger; and in response to the detecting the particular trigger to kernel resume, performing a kernel resume.

[0056] Another example implementation includes a portable playback device comprising: a control interface comprising a power button and transport controls; one or more speakers; one or more amplifiers configured to drive the one or more speakers; a battery; communications interfaces comprising an IEEE 802.11-compatible network interface and an IEEE 802.15-compatible interface; a main system-on-chip (SoC) comprising one or more main processor cores, an auxiliary processor core configured to enable the one or more main processor cores upon receiving an interrupt, and a kernel that executes on the one or more main processor cores, wherein a kernel suspend of the kernel disables the one or more main processor cores; a power management microcontroller configured to perform functions comprising: during kernel suspend of the kernel, monitoring, via one or more sensors of the power management microcontroller, the battery for conditions corresponding to respective wake-on-battery triggers, wherein the wake-on-battery triggers comprise (i) a battery fault, (ii) a critical battery level, and (iii) battery charging initiated; detecting that the monitored conditions correspond to a particular wake-on-battery trigger; and in response to detecting that the monitored conditions correspond to particular wake-on-battery trigger, sending, to the auxiliary processor core, an interrupt corresponding to the particular wake-on-battery trigger, wherein the interrupt causes the auxiliary processor core to enable the one or more main processor cores and resume the kernel from kernel suspend; and a housing carrying the one or more speakers, the one or more amplifiers, the battery, the communications interfaces, the power management microcontroller, and the main SoC, wherein the kernel is configured to perform functions comprising: after resuming from kernel suspend, adding a first kernel resume source event to a power event queue, the first kernel resume source event indicating the particular wake-on-battery trigger, wherein a power coordinator background process is configured to read the first kernel resume source event from the power event queue and send data indicating the particular wake-on-battery trigger to one or more client programs via one or more inter-process communication (IPC) mechanisms.

[0057] While some embodiments described herein may refer to functions performed by given actors, such as “users” and/or other entities, it should be understood that this description is for purposes of explanation only. The claims should not be interpreted to require action by any such example actor unless explicitly required by the language of the claims themselves.

[0058] Moreover, some functions are described herein as being performed “based on” or “in response to” another element or function. “Based on” should be understood that one element or function is related to another function or element. “In response to” should be understood that one element or function is a necessary result of another function or element. For the sake of brevity, functions are generally described as being based on another function when a functional link exists; however, such disclosure should be understood as disclosing either type of functional relationship.

II. Example Operation Environment

[0059] FIG. 1A is a partial cutaway view of a media playback system **100** distributed in an environment **101** (e.g., a house). The media playback system **100** comprises one or more playback devices **110** (identified individually as playback devices **110a-n**), one or more network microphone devices (“NMDs”), **120** (identified individually as NMDs **120a-c**), and one or more control devices **130** (identified individually as control devices **130a** and **130b**).

[0060] As used herein the term “playback device” can generally refer to a network device configured to receive, process, and output data of a media playback system. For example, a playback device can be a network device that receives and processes audio content. In some embodiments, a playback device includes one or more transducers or speakers powered by one or more amplifiers. In other embodiments, however, a playback device includes one of (or neither of) the speaker and the amplifier. For instance, a playback device can comprise one or more amplifiers configured to drive one or more speakers external to the playback device via a corresponding wire or cable.

[0061] Moreover, as used herein the term NMD (i.e., a “network microphone device”) can generally refer to a network device that is configured for audio detection. In some embodiments, an NMD is a stand-alone device configured primarily for audio detection. In other embodiments, an NMD is incorporated into a playback device (or vice versa).

[0062] The term “control device” can generally refer to a network device configured to perform functions relevant to facilitating user access, control, and/or configuration of the media playback system **100**.

[0063] Each of the playback devices **110** is configured to receive audio signals or data from one or more media sources (e.g., one or more remote servers, one or more local devices) and play back the received audio signals or data as sound. The one or more NMDs **120** are configured to receive spoken word commands, and the one or more control devices **130** are configured to receive user input. In response to the received spoken word commands and/or user input, the media playback system **100** can play back audio via one or more of the playback devices **110**.

[0064] In certain embodiments, the playback devices **110** are configured to commence playback of media content in response to a trigger. For instance, one or more of the playback devices **110** can be configured to play back a morning playlist upon detection of an associated trigger condition (e.g., presence of a user in a kitchen, detection of a coffee machine operation). In some embodiments, for example, the media playback system **100** is configured to play back audio from a first playback device (e.g., the playback device **100a**) in synchrony with a second playback device (e.g., the playback device **100b**). Interactions between the playback devices **110**, NMDs **120**, and/or control devices **130** of the media playback system **100** configured in accordance with the various embodiments of the disclosure are described in greater detail below with respect to FIGS. 1B-6.

[0065] In the illustrated embodiment of FIG. 1A, the environment **101** comprises a household having several rooms, spaces, and/or playback zones, including (clockwise from upper left) a master bathroom **101a**, a master bedroom **101b**, a second bedroom **101c**, a family room or den **101d**, an office **101e**, a living room **101f**, a dining room **101g**, a kitchen **101h**, and an outdoor patio

101i. While certain embodiments and examples are described below in the context of a home environment, the technologies described herein may be implemented in other types of environments. In some embodiments, for example, the media playback system **100** can be implemented in one or more commercial settings (e.g., a restaurant, mall, airport, hotel, a retail or other store), one or more vehicles (e.g., a sports utility vehicle, bus, car, a ship, a boat, an airplane), multiple environments (e.g., a combination of home and vehicle environments), and/or another suitable environment where multi-zone audio may be desirable.

[0066] The media playback system **100** can comprise one or more playback zones, some of which may correspond to the rooms in the environment **101**. The media playback system **100** can be established with one or more playback zones, after which additional zones may be added, or removed to form, for example, the configuration shown in FIG. **1A**. Each zone may be given a name according to a different room or space such as the office **101e**, master bathroom **101a**, master bedroom **101b**, the second bedroom **101c**, kitchen **101h**, dining room **101g**, living room **101f**, and/or the balcony **101i**. In some aspects, a single playback zone may include multiple rooms or spaces. In certain aspects, a single room or space may include multiple playback zones.

[0067] In the illustrated embodiment of FIG. **1A**, the master bathroom **101a**, the second bedroom **101c**, the office **101e**, the living room **101f**, the dining room **101g**, the kitchen **101h**, and the outdoor patio **101i** each include one playback device **110**, and the master bedroom **101b** and the den **101d** include a plurality of playback devices **110**. In the master bedroom **101b**, the playback devices **110i** and **110m** may be configured, for example, to play back audio content in synchrony as individual ones of playback devices **110**, as a bonded playback zone, as a consolidated playback device, and/or any combination thereof. Similarly, in the den **101d**, the playback devices **110h-j** can be configured, for instance, to play back audio content in synchrony as individual ones of playback devices **110**, as one or more bonded playback devices, and/or as one or more consolidated playback devices. Additional details regarding bonded and consolidated playback devices are described below with respect to FIGS. **1B** and **1E** and **14-1M**.

[0068] In some aspects, one or more of the playback zones in the environment **101** may each be playing different audio content. For instance, a user may be grilling on the patio **101i** and listening to hip hop music being played by the playback device **110c** while another user is preparing food in the kitchen **101h** and listening to classical music played by the playback device **110b**. In another example, a playback zone may play the same audio content in synchrony with another playback zone. For instance, the user may be in the office **101e** listening to the playback device **110f** playing back the same hip hop music being played back by playback device **110c** on the patio **101i**. In some aspects, the playback devices **110c** and **110f** play back the hip hop music in synchrony such that the user perceives that the audio content is being played seamlessly (or at least substantially seamlessly) while moving between different playback zones. Additional details regarding audio playback synchronization among playback devices and/or zones can be found, for example, in U.S. Pat. No. 8,234,395 entitled, "System and method for synchronizing operations among a plurality of independently clocked digital data processing devices," which is incorporated herein by reference in its entirety.

a. Suitable Media Playback System

[0069] FIG. **1B** is a schematic diagram of the media playback system **100** and a cloud network **102**. For ease of illustration, certain devices of the media playback system **100** and the cloud network **102** are omitted from FIG. **1B**. One or more communication links **103** (referred to hereinafter as "the links **103**") communicatively couple the media playback system **100** and the cloud network **102**.

[0070] The links **103** can comprise, for example, one or more wired networks, one or more wireless networks, one or more wide area networks (WAN), one or more local area networks (LAN), one or more personal area networks (PAN), one or more telecommunication networks (e.g., one or more Global System for Mobiles (GSM) networks, Code Division Multiple Access (CDMA) networks,

Long-Term Evolution (LTE) networks, 5G communication network networks, and/or other suitable data transmission protocol networks), etc. The cloud network **102** is configured to deliver media content (e.g., audio content, video content, photographs, social media content) to the media playback system **100** in response to a request transmitted from the media playback system **100** via the links **103**. In some embodiments, the cloud network **102** is further configured to receive data (e.g. voice input data) from the media playback system **100** and correspondingly transmit commands and/or media content to the media playback system **100**.

[0071] The cloud network **102** comprises computing devices **106** (identified separately as a first computing device **106a**, a second computing device **106b**, and a third computing device **106c**). The computing devices **106** can comprise individual computers or servers, such as, for example, a media streaming service server storing audio and/or other media content, a voice service server, a social media server, a media playback system control server, etc. In some embodiments, one or more of the computing devices **106** comprise modules of a single computer or server. In certain embodiments, one or more of the computing devices **106** comprise one or more modules, computers, and/or servers. Moreover, while the cloud network **102** is described above in the context of a single cloud network, in some embodiments the cloud network **102** comprises a plurality of cloud networks comprising communicatively coupled computing devices. Furthermore, while the cloud network **102** is shown in FIG. **1B** as having three of the computing devices **106**, in some embodiments, the cloud network **102** comprises fewer (or more than) three computing devices **106**.

[0072] The media playback system **100** is configured to receive media content from the networks **102** via the links **103**. The received media content can comprise, for example, a Uniform Resource Identifier (URI) and/or a Uniform Resource Locator (URL). For instance, in some examples, the media playback system **100** can stream, download, or otherwise obtain data from a URI or a URL corresponding to the received media content. A network **104** communicatively couples the links **103** and at least a portion of the devices (e.g., one or more of the playback devices **110**, NMDs **120**, and/or control devices **130**) of the media playback system **100**. The network **104** can include, for example, a wireless network (e.g., a Wi-Fi network, a Bluetooth, a Z-Wave network, a ZigBee, and/or other suitable wireless communication protocol network) and/or a wired network (e.g., a network comprising Ethernet, Universal Serial Bus (USB), and/or another suitable wired communication). As those of ordinary skill in the art will appreciate, as used herein, “Wi-Fi” can refer to several different communication protocols including, for example, Institute of Electrical and Electronics Engineers (IEEE) 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac, 802.11ad, 802.11af, 802.11ah, 802.11ai, 802.11aj, 802.11aq, 802.11ax, 802.11ay, 802.15, etc. transmitted at 2.4 Gigahertz (GHz), 5 GHz, and/or another suitable frequency.

[0073] In some embodiments, the network **104** comprises a dedicated communication network that the media playback system **100** uses to transmit messages between individual devices and/or to transmit media content to and from media content sources (e.g., one or more of the computing devices **106**). In certain embodiments, the network **104** is configured to be accessible only to devices in the media playback system **100**, thereby reducing interference and competition with other household devices. In other embodiments, however, the network **104** comprises an existing household communication network (e.g., a household Wi-Fi network). In some embodiments, the links **103** and the network **104** comprise one or more of the same networks. In some aspects, for example, the links **103** and the network **104** comprise a telecommunication network (e.g., an LTE network, a 5G network). Moreover, in some embodiments, the media playback system **100** is implemented without the network **104**, and devices comprising the media playback system **100** can communicate with each other, for example, via one or more direct connections, PANs, telecommunication networks, and/or other suitable communication links.

[0074] In some embodiments, audio content sources may be regularly added or removed from the media playback system **100**. In some embodiments, for example, the media playback system **100**

performs an indexing of media items when one or more media content sources are updated, added to, and/or removed from the media playback system **100**. The media playback system **100** can scan identifiable media items in some or all folders and/or directories accessible to the playback devices **110**, and generate or update a media content database comprising metadata (e.g., title, artist, album, track length) and other associated information (e.g., URIs, URLs) for each identifiable media item found. In some embodiments, for example, the media content database is stored on one or more of the playback devices **110**, network microphone devices **120**, and/or control devices **130**.

[0075] In the illustrated embodiment of FIG. **1B**, the playback devices **110l** and **110m** comprise a group **107a**. The playback devices **110a** and **110m** can be positioned in different rooms in a household and be grouped together in the group **107a** on a temporary or permanent basis based on user input received at the control device **130a** and/or another control device **130** in the media playback system **100**. When arranged in the group **107a**, the playback devices **110l** and **110m** can be configured to play back the same or similar audio content in synchrony from one or more audio content sources. In certain embodiments, for example, the group **107a** comprises a bonded zone in which the playback devices **110l** and **110m** comprise left audio and right audio channels, respectively, of multi-channel audio content, thereby producing or enhancing a stereo effect of the audio content. In some embodiments, the group **107a** includes additional playback devices **110**. In other embodiments, however, the media playback system **100** omits the group **107a** and/or other grouped arrangements of the playback devices **110**. Additional details regarding groups and other arrangements of playback devices are described in further detail below with respect to FIGS. **1-I** through **IM**.

[0076] The media playback system **100** includes the NMDs **120a** and **120d**, each comprising one or more microphones configured to receive voice utterances from a user. In the illustrated embodiment of FIG. **1B**, the NMD **120a** is a standalone device and the NMD **120d** is integrated into the playback device **110n**. The NMD **120a**, for example, is configured to receive voice input **121** from a user **123**. In some embodiments, the NMD **120a** transmits data associated with the received voice input **121** to a voice assistant service (VAS) configured to (i) process the received voice input data and (ii) transmit a corresponding command to the media playback system **100**. In some aspects, for example, the computing device **106c** comprises one or more modules and/or servers of a VAS (e.g., a VAS operated by one or more of SONOS®, AMAZON®, GOOGLE®, APPLE®, MICROSOFT®). The computing device **106c** can receive the voice input data from the NMD **120a** via the network **104** and the links **103**. In response to receiving the voice input data, the computing device **106c** processes the voice input data (i.e., “Play Hey Jude by The Beatles”), and determines that the processed voice input includes a command to play a song (e.g., “Hey Jude”). The computing device **106c** accordingly transmits commands to the media playback system **100** to play back “Hey Jude” by the Beatles from a suitable media service (e.g., via one or more of the computing devices **106**) on one or more of the playback devices **110**.

b. Suitable Playback Devices

[0077] FIG. **1C** is a block diagram of the playback device **110a** comprising an input/output **111**. The input/output **111** can include an analog I/O **111a** (e.g., one or more wires, cables, and/or other suitable communication links configured to carry analog signals) and/or a digital I/O **111b** (e.g., one or more wires, cables, or other suitable communication links configured to carry digital signals). In some embodiments, the analog I/O **111a** is an audio line-in input connection comprising, for example, an auto-detecting 3.5 mm audio line-in connection. In some embodiments, the digital I/O **111b** comprises a Sony/Philips Digital Interface Format (S/PDIF) communication interface and/or cable and/or a Toshiba Link (TOSLINK) cable. In some embodiments, the digital I/O **111b** comprises a High-Definition Multimedia Interface (HDMI) interface and/or cable. In some embodiments, the digital I/O **111b** includes one or more wireless communication links comprising, for example, a radio frequency (RF), infrared, Wi-Fi, Bluetooth, or another suitable communication protocol. In certain embodiments, the analog I/O **111a** and the

digital **111b** comprise interfaces (e.g., ports, plugs, jacks) configured to receive connectors of cables transmitting analog and digital signals, respectively, without necessarily including cables. [0078] The playback device **110a**, for example, can receive media content (e.g., audio content comprising music and/or other sounds) from a local audio source **105** via the input/output **111** (e.g., a cable, a wire, a PAN, a Bluetooth connection, an ad hoc wired or wireless communication network, and/or another suitable communication link). The local audio source **105** can comprise, for example, a mobile device (e.g., a smartphone, a tablet, a laptop computer) or another suitable audio component (e.g., a television, a desktop computer, an amplifier, a phonograph, a Blu-ray player, a memory storing digital media files). In some aspects, the local audio source **105** includes local music libraries on a smartphone, a computer, a networked-attached storage (NAS), and/or another suitable device configured to store media files. In certain embodiments, one or more of the playback devices **110**, NMDs **120**, and/or control devices **130** comprise the local audio source **105**. In other embodiments, however, the media playback system omits the local audio source **105** altogether. In some embodiments, the playback device **110a** does not include an input/output **111** and receives all audio content via the network **104**.

[0079] The playback device **110a** further comprises electronics **112**, a user interface **113** (e.g., one or more buttons, knobs, dials, touch-sensitive surfaces, displays, touchscreens), and one or more transducers **114** (referred to hereinafter as “the transducers **114**”). The electronics **112** is configured to receive audio from an audio source (e.g., the local audio source **105**) via the input/output **111**, one or more of the computing devices **106a-c** via the network **104** (FIG. **1B**)), amplify the received audio, and output the amplified audio for playback via one or more of the transducers **114**. In some embodiments, the playback device **110a** optionally includes one or more microphones **115** (e.g., a single microphone, a plurality of microphones, a microphone array) (hereinafter referred to as “the microphones **115**”). In certain embodiments, for example, the playback device **110a** having one or more of the optional microphones **115** can operate as an NMD configured to receive voice input from a user and correspondingly perform one or more operations based on the received voice input.

[0080] In the illustrated embodiment of FIG. **1C**, the electronics **112** comprise one or more processors **112a** (referred to hereinafter as “the processors **112a**”), memory **112b**, software components **112c**, a network interface **112d**, one or more audio processing components **112g** (referred to hereinafter as “the audio components **112g**”), one or more audio amplifiers **112h** (referred to hereinafter as “the amplifiers **112h**”), and power **112i** (e.g., one or more power supplies, power cables, power receptacles, batteries, induction coils, Power-over Ethernet (POE) interfaces, and/or other suitable sources of electric power). In some embodiments, the electronics **112** optionally include one or more other components **112j** (e.g., one or more sensors, video displays, touchscreens, battery charging bases).

[0081] The processors **112a** can comprise clock-driven computing component(s) configured to process data, and the memory **112b** can comprise a computer-readable medium (e.g., a tangible, non-transitory computer-readable medium, data storage loaded with one or more of the software components **112c**) configured to store instructions for performing various operations and/or functions. The processors **112a** are configured to execute the instructions stored on the memory **112b** to perform one or more of the operations. The operations can include, for example, causing the playback device **110a** to retrieve audio data from an audio source (e.g., one or more of the computing devices **106a-c** (FIG. **1B**)), and/or another one of the playback devices **110**. In some embodiments, the operations further include causing the playback device **110a** to send audio data to another one of the playback devices **110a** and/or another device (e.g., one of the NMDs **120**). Certain embodiments include operations causing the playback device **110a** to pair with another of the one or more playback devices **110** to enable a multi-channel audio environment (e.g., a stereo pair, a bonded zone).

[0082] The processors **112a** can be further configured to perform operations causing the playback device **110a** to synchronize playback of audio content with another of the one or more playback

devices **110**. As those of ordinary skill in the art will appreciate, during synchronous playback of audio content on a plurality of playback devices, a listener will preferably be unable to perceive time-delay differences between playback of the audio content by the playback device **110a** and the other one or more other playback devices **110**. Additional details regarding audio playback synchronization among playback devices can be found, for example, in U.S. Pat. No. 8,234,395, which was incorporated by reference above.

[0083] In some embodiments, the memory **112b** is further configured to store data associated with the playback device **110a**, such as one or more zones and/or zone groups of which the playback device **110a** is a member, audio sources accessible to the playback device **110a**, and/or a playback queue that the playback device **110a** (and/or another of the one or more playback devices) can be associated with. The stored data can comprise one or more state variables that are periodically updated and used to describe a state of the playback device **110a**. The memory **112b** can also include data associated with a state of one or more of the other devices (e.g., the playback devices **110**, NMDs **120**, control devices **130**) of the media playback system **100**. In some aspects, for example, the state data is shared during predetermined intervals of time (e.g., every 5 seconds, every 10 seconds, every 60 seconds) among at least a portion of the devices of the media playback system **100**, so that one or more of the devices have the most recent data associated with the media playback system **100**.

[0084] The network interface **112d** is configured to facilitate a transmission of data between the playback device **110a** and one or more other devices on a data network such as, for example, the links **103** and/or the network **104** (FIG. 1B). The network interface **112d** is configured to transmit and receive data corresponding to media content (e.g., audio content, video content, text, photographs) and other signals (e.g., non-transitory signals) comprising digital packet data including an Internet Protocol (IP)-based source address and/or an IP-based destination address. The network interface **112d** can parse the digital packet data such that the electronics **112** properly receives and processes the data destined for the playback device **110a**.

[0085] In the illustrated embodiment of FIG. 1C, the network interface **112d** comprises one or more wireless interfaces **112e** (referred to hereinafter as “the wireless interface **112e**”). The wireless interface **112e** (e.g., a suitable interface comprising one or more antennae) can be configured to wirelessly communicate with one or more other devices (e.g., one or more of the other playback devices **110**, NMDs **120**, and/or control devices **130**) that are communicatively coupled to the network **104** (FIG. 1B) in accordance with a suitable wireless communication protocol (e.g., Wi-Fi, Bluetooth, LTE). In some embodiments, the network interface **112d** optionally includes a wired interface **112f** (e.g., an interface or receptacle configured to receive a network cable such as an Ethernet, a USB-A, USB-C, and/or Thunderbolt cable) configured to communicate over a wired connection with other devices in accordance with a suitable wired communication protocol. In certain embodiments, the network interface **112d** includes the wired interface **112f** and excludes the wireless interface **112e**. In some embodiments, the electronics **112** excludes the network interface **112d** altogether and transmits and receives media content and/or other data via another communication path (e.g., the input/output **111**).

[0086] The audio components **112g** are configured to process and/or filter data comprising media content received by the electronics **112** (e.g., via the input/output **111** and/or the network interface **112d**) to produce output audio signals. In some embodiments, the audio processing components **112g** comprise, for example, one or more digital-to-analog converters (DAC), audio preprocessing components, audio enhancement components, a digital signal processors (DSPs), and/or other suitable audio processing components, modules, circuits, etc. In certain embodiments, one or more of the audio processing components **112g** can comprise one or more subcomponents of the processors **112a**. In some embodiments, the electronics **112** omits the audio processing components **112g**. In some aspects, for example, the processors **112a** execute instructions stored on the memory **112b** to perform audio processing operations to produce the output audio signals.

[0087] The amplifiers **112h** are configured to receive and amplify the audio output signals produced by the audio processing components **112g** and/or the processors **112a**. The amplifiers **112h** can comprise electronic devices and/or components configured to amplify audio signals to levels sufficient for driving one or more of the transducers **114**. In some embodiments, for example, the amplifiers **112h** include one or more switching or class-D power amplifiers. In other embodiments, however, the amplifiers include one or more other types of power amplifiers (e.g., linear gain power amplifiers, class-A amplifiers, class-B amplifiers, class-AB amplifiers, class-C amplifiers, class-D amplifiers, class-E amplifiers, class-F amplifiers, class-G and/or class H amplifiers, and/or another suitable type of power amplifier). In certain embodiments, the amplifiers **112h** comprise a suitable combination of two or more of the foregoing types of power amplifiers. Moreover, in some embodiments, individual ones of the amplifiers **112h** correspond to individual ones of the transducers **114**. In other embodiments, however, the electronics **112** includes a single one of the amplifiers **112h** configured to output amplified audio signals to a plurality of the transducers **114**. In some other embodiments, the electronics **112** omits the amplifiers **112h**.

[0088] The transducers **114** (e.g., one or more speakers and/or speaker drivers) receive the amplified audio signals from the amplifier **112h** and render or output the amplified audio signals as sound (e.g., audible sound waves having a frequency between about 20 Hertz (Hz) and 20 kilohertz (kHz)). In some embodiments, the transducers **114** can comprise a single transducer. In other embodiments, however, the transducers **114** comprise a plurality of audio transducers. In some embodiments, the transducers **114** comprise more than one type of transducer. For example, the transducers **114** can include one or more low frequency transducers (e.g., subwoofers, woofers), mid-range frequency transducers (e.g., mid-range transducers, mid-woofers), and one or more high frequency transducers (e.g., one or more tweeters). As used herein, “low frequency” can generally refer to audible frequencies below about 500 Hz, “mid-range frequency” can generally refer to audible frequencies between about 500 Hz and about 2 kHz, and “high frequency” can generally refer to audible frequencies above 2 kHz. In certain embodiments, however, one or more of the transducers **114** comprise transducers that do not adhere to the foregoing frequency ranges. For example, one of the transducers **114** may comprise a mid-woofer transducer configured to output sound at frequencies between about 200 Hz and about 5 kHz.

[0089] By way of illustration, SONOS, Inc. presently offers (or has offered) for sale certain playback devices including, for example, a “SONOS ONE,” “PLAY: 1,” “PLAY: 3,” “PLAY: 5,” “PLAYBAR,” “PLAYBASE,” “CONNECT: AMP,” “CONNECT,” and “SUB.” Other suitable playback devices may additionally or alternatively be used to implement the playback devices of example embodiments disclosed herein. Additionally, one of ordinary skilled in the art will appreciate that a playback device is not limited to the examples described herein or to SONOS product offerings. In some embodiments, for example, one or more playback devices **110** comprises wired or wireless headphones (e.g., over-the-ear headphones, on-ear headphones, in-ear earphones). In other embodiments, one or more of the playback devices **110** comprise a docking station and/or an interface configured to interact with a docking station for personal mobile media playback devices. In certain embodiments, a playback device may be integral to another device or component such as a television, a lighting fixture, or some other device for indoor or outdoor use. In some embodiments, a playback device omits a user interface and/or one or more transducers. For example, FIG. 1D is a block diagram of a playback device **110p** comprising the input/output **111** and electronics **112** without the user interface **113** or transducers **114**.

[0090] FIG. 1E is a block diagram of a bonded playback device **110q** comprising the playback device **110a** (FIG. 1C) sonically bonded with the playback device **110i** (e.g., a subwoofer) (FIG. 1A). In the illustrated embodiment, the playback devices **110a** and **110i** are separate ones of the playback devices **110** housed in separate enclosures. In some embodiments, however, the bonded playback device **110q** comprises a single enclosure housing both the playback devices **110a** and **110i**. The bonded playback device **110q** can be configured to process and reproduce sound

differently than an unbonded playback device (e.g., the playback device **110a** of FIG. 1C) and/or paired or bonded playback devices (e.g., the playback devices **110l** and **110m** of FIG. 1B). In some embodiments, for example, the playback device **110a** is full-range playback device configured to render low frequency, mid-range frequency, and high frequency audio content, and the playback device **110i** is a subwoofer configured to render low frequency audio content. In some aspects, the playback device **110a**, when bonded with the first playback device, is configured to render only the mid-range and high frequency components of a particular audio content, while the playback device **110i** renders the low frequency component of the particular audio content. In some embodiments, the bonded playback device **110q** includes additional playback devices and/or another bonded playback device. Additional playback device embodiments are described in further detail below with respect to FIGS. 2A-3D.

c. Suitable Network Microphone Devices (NMDs)

[0091] FIG. 1F is a block diagram of the NMD **120a** (FIGS. 1A and 1B). The NMD **120a** includes one or more voice processing components **124** (hereinafter “the voice components **124**”) and several components described with respect to the playback device **110a** (FIG. 1C) including the processors **112a**, the memory **112b**, and the microphones **115**. The NMD **120a** optionally comprises other components also included in the playback device **110a** (FIG. 1C), such as the user interface **113** and/or the transducers **114**. In some embodiments, the NMD **120a** is configured as a media playback device (e.g., one or more of the playback devices **110**), and further includes, for example, one or more of the audio components **112g** (FIG. 1C), the amplifiers **114**, and/or other playback device components. In certain embodiments, the NMD **120a** comprises an Internet of Things (IoT) device such as, for example, a thermostat, alarm panel, fire and/or smoke detector, etc. In some embodiments, the NMD **120a** comprises the microphones **115**, the voice processing **124**, and only a portion of the components of the electronics **112** described above with respect to FIG. 1B. In some aspects, for example, the NMD **120a** includes the processor **112a** and the memory **112b** (FIG. 1B), while omitting one or more other components of the electronics **112**. In some embodiments, the NMD **120a** includes additional components (e.g., one or more sensors, cameras, thermometers, barometers, hygrometers).

[0092] In some embodiments, an NMD can be integrated into a playback device. FIG. 1G is a block diagram of a playback device **110r** comprising an NMD **120d**. The playback device **110r** can comprise many or all of the components of the playback device **110a** and further include the microphones **115** and voice processing **124** (FIG. 1F). The playback device **110r** optionally includes an integrated control device **130c**. The control device **130c** can comprise, for example, a user interface (e.g., the user interface **113** of FIG. 1B) configured to receive user input (e.g., touch input, voice input) without a separate control device. In other embodiments, however, the playback device **110r** receives commands from another control device (e.g., the control device **130a** of FIG. 1B).

[0093] Referring again to FIG. 1F, the microphones **115** are configured to acquire, capture, and/or receive sound from an environment (e.g., the environment **101** of FIG. 1A) and/or a room in which the NMD **120a** is positioned. The received sound can include, for example, vocal utterances, audio played back by the NMD **120a** and/or another playback device, background voices, ambient sounds, etc. The microphones **115** convert the received sound into electrical signals to produce microphone data. The voice processing **124** receives and analyzes the microphone data to determine whether a voice input is present in the microphone data. The voice input can comprise, for example, a wake word followed by an utterance including a user request. As those of ordinary skill in the art will appreciate, an wake word is a word or other audio cue that signifying a user voice input. For instance, in querying the AMAZON® VAS, a user might speak the wake word “Alexa.” Other examples include “Ok, Google” for invoking the GOOGLE® VAS and “Hey, Siri” for invoking the APPLE® VAS.

[0094] After detecting the wake word, voice processing **124** monitors the microphone data for an

accompanying user request in the voice input. The user request may include, for example, a command to control a third-party device, such as a thermostat (e.g., NEST® thermostat), an illumination device (e.g., a PHILIPS HUE® lighting device), or a media playback device (e.g., a Sonos® playback device). For example, a user might speak the wake word “Alexa” followed by the utterance “set the thermostat to **68** degrees” to set a temperature in a home (e.g., the environment **101** of FIG. **1A**). The user might speak the same wake word followed by the utterance “turn on the living room” to turn on illumination devices in a living room area of the home. The user may similarly speak an wake word followed by a request to play a particular song, an album, or a playlist of music on a playback device in the home.

d. Suitable Control Devices

[0095] FIG. **1H** is a partially schematic diagram of the control device **130a** (FIGS. **1A** and **1B**). As used herein, the term “control device” can be used interchangeably with “controller” or “control system.” Among other features, the control device **130a** is configured to receive user input related to the media playback system **100** and, in response, cause one or more devices in the media playback system **100** to perform an action(s) or operation(s) corresponding to the user input. In the illustrated embodiment, the control device **130a** comprises a smartphone (e.g., an iPhone™, an Android phone) on which media playback system controller application software is installed. In some embodiments, the control device **130a** comprises, for example, a tablet (e.g., an iPad™), a computer (e.g., a laptop computer, a desktop computer), and/or another suitable device (e.g., a television, an automobile audio head unit, an IoT device). In certain embodiments, the control device **130a** comprises a dedicated controller for the media playback system **100**. In other embodiments, as described above with respect to FIG. **1G**, the control device **130a** is integrated into another device in the media playback system **100** (e.g., one more of the playback devices **110**, NMDs **120**, and/or other suitable devices configured to communicate over a network).

[0096] The control device **130a** includes electronics **132**, a user interface **133**, one or more speakers **134**, and one or more microphones **135**. The electronics **132** comprise one or more processors **132a** (referred to hereinafter as “the processors **132 a**”), a memory **132b**, software components **132c**, and a network interface **132d**. The processor **132a** can be configured to perform functions relevant to facilitating user access, control, and configuration of the media playback system **100**. The memory **132b** can comprise data storage that can be loaded with one or more of the software components executable by the processor **302** to perform those functions. The software components **132c** can comprise applications and/or other executable software configured to facilitate control of the media playback system **100**. The memory **112b** can be configured to store, for example, the software components **132c**, media playback system controller application software, and/or other data associated with the media playback system **100** and the user.

[0097] The network interface **132d** is configured to facilitate network communications between the control device **130a** and one or more other devices in the media playback system **100**, and/or one or more remote devices. In some embodiments, the network interface **132** is configured to operate according to one or more suitable communication industry standards (e.g., infrared, radio, wired standards including IEEE 802.3, wireless standards including IEEE 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac, 802.15, 4G, LTE). The network interface **132d** can be configured, for example, to transmit data to and/or receive data from the playback devices **110**, the NMDs **120**, other ones of the control devices **130**, one of the computing devices **106** of FIG. **1B**, devices comprising one or more other media playback systems, etc. The transmitted and/or received data can include, for example, playback device control commands, state variables, playback zone and/or zone group configurations. For instance, based on user input received at the user interface **133**, the network interface **132d** can transmit a playback device control command (e.g., volume control, audio playback control, audio content selection) from the control device **304** to one or more of the playback devices **100**. The network interface **132d** can also transmit and/or receive configuration changes such as, for example, adding/removing one or more playback devices **100** to/from a zone,

adding/removing one or more zones to/from a zone group, forming a bonded or consolidated player, separating one or more playback devices from a bonded or consolidated player, among others. Additional description of zones and groups can be found below with respect to FIGS. 1-1 through 1M.

[0098] The user interface **133** is configured to receive user input and can facilitate ‘control of the media playback system **100**. The user interface **133** includes media content art **133a** (e.g., album art, lyrics, videos), a playback status indicator **133b** (e.g., an elapsed and/or remaining time indicator), media content information region **133c**, a playback control region **133d**, and a zone indicator **133e**. The media content information region **133c** can include a display of relevant information (e.g., title, artist, album, genre, release year) about media content currently playing and/or media content in a queue or playlist. The playback control region **133d** can include selectable (e.g., via touch input and/or via a cursor or another suitable selector) icons to cause one or more playback devices in a selected playback zone or zone group to perform playback actions such as, for example, play or pause, fast forward, rewind, skip to next, skip to previous, enter/exit shuffle mode, enter/exit repeat mode, enter/exit cross fade mode, etc. The playback control region **133d** may also include selectable icons to modify equalization settings, playback volume, and/or other suitable playback actions. In the illustrated embodiment, the user interface **133** comprises a display presented on a touch screen interface of a smartphone (e.g., an iPhone™, an Android phone). In some embodiments, however, user interfaces of varying formats, styles, and interactive sequences may alternatively be implemented on one or more network devices to provide comparable control access to a media playback system.

[0099] The one or more speakers **134** (e.g., one or more transducers) can be configured to output sound to the user of the control device **130a**. In some embodiments, the one or more speakers comprise individual transducers configured to correspondingly output low frequencies, mid-range frequencies, and/or high frequencies. In some aspects, for example, the control device **130a** is configured as a playback device (e.g., one of the playback devices **110**). Similarly, in some embodiments the control device **130a** is configured as an NMD (e.g., one of the NMDs **120**), receiving voice commands and other sounds via the one or more microphones **135**.

[0100] The one or more microphones **135** can comprise, for example, one or more condenser microphones, electret condenser microphones, dynamic microphones, and/or other suitable types of microphones or transducers. In some embodiments, two or more of the microphones **135** are arranged to capture location information of an audio source (e.g., voice, audible sound) and/or configured to facilitate filtering of background noise. Moreover, in certain embodiments, the control device **130a** is configured to operate as playback device and an NMD. In other embodiments, however, the control device **130a** omits the one or more speakers **134** and/or the one or more microphones **135**. For instance, the control device **130a** may comprise a device (e.g., a thermostat, an IoT device, a network device) comprising a portion of the electronics **132** and the user interface **133** (e.g., a touch screen) without any speakers or microphones. Additional control device embodiments are described in further detail below with respect to FIGS. 4A-4D and 5.

e. Suitable Playback Device Configurations

[0101] FIGS. 1-1 through 1M show example configurations of playback devices in zones and zone groups. Referring first to FIG. 1M, in one example, a single playback device may belong to a zone. For example, the playback device **110g** in the second bedroom **101c** (FIG. 1A) may belong to Zone C. In some implementations described below, multiple playback devices may be “bonded” to form a “bonded pair” which together form a single zone. For example, the playback device **110l** (e.g., a left playback device) can be bonded to the playback device **110m** (e.g., a right playback device) to form Zone A. Bonded playback devices may have different playback responsibilities (e.g., channel responsibilities). In another implementation described below, multiple playback devices may be merged to form a single zone. For example, the playback device **110h** (e.g., a front playback device) may be merged with the playback device **110i** (e.g., a subwoofer), and the playback devices

110j and **110k** (e.g., left and right surround speakers, respectively) to form a single Zone D. In another example, the playback devices **110g** and **110h** can be merged to form a merged group or a zone group **108b**. The merged playback devices **110g** and **110h** may not be specifically assigned different playback responsibilities. That is, the merged playback devices **110h** and **110i** may, aside from playing audio content in synchrony, each play audio content as they would if they were not merged.

[0102] Each zone in the media playback system **100** may be provided for control as a single user interface (UI) entity. For example, Zone A may be provided as a single entity named Master Bathroom. Zone B may be provided as a single entity named Master Bedroom. Zone C may be provided as a single entity named Second Bedroom.

[0103] Playback devices that are bonded may have different playback responsibilities, such as responsibilities for certain audio channels. For example, as shown in FIG. 1-I, the playback devices **110l** and **110m** may be bonded so as to produce or enhance a stereo effect of audio content. In this example, the playback device **110l** may be configured to play a left channel audio component, while the playback device **110k** may be configured to play a right channel audio component. In some implementations, such stereo bonding may be referred to as “pairing.”

[0104] Additionally, bonded playback devices may have additional and/or different respective speaker drivers. As shown in FIG. 1J, the playback device **110h** named Front may be bonded with the playback device **110i** named SUB. The Front device **110h** can be configured to render a range of mid to high frequencies and the SUB device **110i** can be configured render low frequencies. When unbonded, however, the Front device **110h** can be configured render a full range of frequencies. As another example, FIG. 1K shows the Front and SUB devices **110h** and **110i** further bonded with Left and Right playback devices **110j** and **110k**, respectively. In some implementations, the Right and Left devices **110j** and **110k** can be configured to form surround or “satellite” channels of a home theater system. The bonded playback devices **110h**, **110i**, **110j**, and **110k** may form a single Zone D (FIG. 1M).

[0105] Playback devices that are merged may not have assigned playback responsibilities, and may each render the full range of audio content the respective playback device is capable of. Nevertheless, merged devices may be represented as a single UI entity (i.e., a zone, as discussed above). For instance, the playback devices **110a** and **110n** the master bathroom have the single UI entity of Zone A. In one embodiment, the playback devices **110a** and **110n** may each output the full range of audio content each respective playback devices **110a** and **110n** are capable of, in synchrony.

[0106] In some embodiments, an NMD is bonded or merged with another device so as to form a zone. For example, the NMD **120b** may be bonded with the playback device **110e**, which together form Zone F, named Living Room. In other embodiments, a stand-alone network microphone device may be in a zone by itself. In other embodiments, however, a stand-alone network microphone device may not be associated with a zone. Additional details regarding associating network microphone devices and playback devices as designated or default devices may be found, for example, in previously referenced U.S. patent application Ser. No. 15/438,749.

[0107] Zones of individual, bonded, and/or merged devices may be grouped to form a zone group. For example, referring to FIG. 1M, Zone A may be grouped with Zone B to form a zone group **108a** that includes the two zones. Similarly, Zone G may be grouped with Zone H to form the zone group **108b**. As another example, Zone A may be grouped with one or more other Zones C-I. The Zones A-I may be grouped and ungrouped in numerous ways. For example, three, four, five, or more (e.g., all) of the Zones A-I may be grouped. When grouped, the zones of individual and/or bonded playback devices may play back audio in synchrony with one another, as described in previously referenced U.S. Pat. No. 8,234,395. Playback devices may be dynamically grouped and ungrouped to form new or different groups that synchronously play back audio content.

[0108] In various implementations, the zones in an environment may be the default name of a zone

within the group or a combination of the names of the zones within a zone group. For example, Zone Group **108b** can have be assigned a name such as “Dining+Kitchen”, as shown in FIG. **1M**. In some embodiments, a zone group may be given a unique name selected by a user.

[0109] Certain data may be stored in a memory of a playback device (e.g., the memory **112c** of FIG. **1C**) as one or more state variables that are periodically updated and used to describe the state of a playback zone, the playback device(s), and/or a zone group associated therewith. The memory may also include the data associated with the state of the other devices of the media system, and shared from time to time among the devices so that one or more of the devices have the most recent data associated with the system.

[0110] In some embodiments, the memory may store instances of various variable types associated with the states. Variables instances may be stored with identifiers (e.g., tags) corresponding to type. For example, certain identifiers may be a first type “a1” to identify playback device(s) of a zone, a second type “b1” to identify playback device(s) that may be bonded in the zone, and a third type “c1” to identify a zone group to which the zone may belong. As a related example, identifiers associated with the second bedroom **101c** may indicate that the playback device is the only playback device of the Zone C and not in a zone group. Identifiers associated with the Den may indicate that the Den is not grouped with other zones but includes bonded playback devices **110h-110k**. Identifiers associated with the Dining Room may indicate that the Dining Room is part of the Dining+Kitchen zone group **108b** and that devices **110b** and **110d** are grouped (FIG. **1L**). Identifiers associated with the Kitchen may indicate the same or similar information by virtue of the Kitchen being part of the Dining+Kitchen zone group **108b**. Other example zone variables and identifiers are described below.

[0111] In yet another example, the media playback system **100** may variables or identifiers representing other associations of zones and zone groups, such as identifiers associated with Areas, as shown in FIG. **1M**. An area may involve a cluster of zone groups and/or zones not within a zone group. For instance, FIG. **1M** shows an Upper Area **109a** including Zones A-D, and a Lower Area **109b** including Zones E-I. In one aspect, an Area may be used to invoke a cluster of zone groups and/or zones that share one or more zones and/or zone groups of another cluster. In another aspect, this differs from a zone group, which does not share a zone with another zone group. Further examples of techniques for implementing Areas may be found, for example, in U.S. application Ser. No. 15/682,506 filed Aug. 21, 2017 and titled “Room Association Based on Name,” and U.S. Pat. No. 8,483,853 filed Sep. 11, 2007, and titled “Controlling and manipulating groupings in a multi-zone media system.” Each of these applications is incorporated herein by reference in its entirety. In some embodiments, the media playback system **100** may not implement Areas, in which case the system may not store variables associated with Areas.

III. Example Systems and Devices

[0112] FIG. **2A** is a front isometric view of a playback device **210** configured in accordance with aspects of the disclosed technology. FIG. **2B** is a front isometric view of the playback device **210** without a grille **216e**. FIG. **2C** is an exploded view of the playback device **210**. Referring to FIGS. **2A-2C** together, the playback device **210** comprises a housing **216** that includes an upper portion **216a**, a right or first side portion **216b**, a lower portion **216c**, a left or second side portion **216d**, the grille **216e**, and a rear portion **216f**. A plurality of fasteners **216g** (e.g., one or more screws, rivets, clips) attaches a frame **216h** to the housing **216**. A cavity **216j** (FIG. **2C**) in the housing **216** is configured to receive the frame **216h** and electronics **212**. The frame **216h** is configured to carry a plurality of transducers **214** (identified individually in FIG. **2B** as transducers **214a-f**). The electronics **212** (e.g., the electronics **112** of FIG. **1C**) is configured to receive audio content from an audio source and send electrical signals corresponding to the audio content to the transducers **214** for playback.

[0113] The transducers **214** are configured to receive the electrical signals from the electronics **112**, and further configured to convert the received electrical signals into audible sound during playback.

For instance, the transducers **214a-c** (e.g., tweeters) can be configured to output high frequency sound (e.g., sound waves having a frequency greater than about 2 kHz). The transducers **214d-f** (e.g., mid-woofers, woofers, midrange speakers) can be configured output sound at frequencies lower than the transducers **214a-c** (e.g., sound waves having a frequency lower than about 2 kHz). In some embodiments, the playback device **210** includes a number of transducers different than those illustrated in FIGS. **2A-2C**. For example, the playback device **210** can include fewer than six transducers (e.g., one, two, three). In other embodiments, however, the playback device **210** includes more than six transducers (e.g., nine, ten). Moreover, in some embodiments, all or a portion of the transducers **214** are configured to operate as a phased array to desirably adjust (e.g., narrow or widen) a radiation pattern of the transducers **214**, thereby altering a user's perception of the sound emitted from the playback device **210**.

[0114] In the illustrated embodiment of FIGS. **2A-2C**, a filter **216i** is axially aligned with the transducer **214b**. The filter **216i** can be configured to desirably attenuate a predetermined range of frequencies that the transducer **214b** outputs to improve sound quality and a perceived sound stage output collectively by the transducers **214**. In some embodiments, however, the playback device **210** omits the filter **216i**. In other embodiments, the playback device **210** includes one or more additional filters aligned with the transducers **214b** and/or at least another of the transducers **214**.

[0115] FIG. **3A** is a block diagram of a portable playback device **310** configured in accordance with embodiments of the disclosed technology. In contrast to the playback device(s) **110**, the portable playback device **310** is configured to facilitate portable (battery-powered) operation in addition to operation while connected to AC (wall) power. The portable playback device **310** includes similar components as the playback device **110a** (FIG. **1C**) including an input/output **311**, electronics **312**, user interface **313**, and transducers **314**, which are similar to the corresponding input/output **111**, electronics **112**, user interface **113**, and transducers **114** of the playback device **110a**, including similar sub-components.

[0116] Yet, the components of the portable playback device **310** may differ in various ways to facilitate portable operation. For instance, the power **312i** may include batteries to power operation of the portable playback device **310** while disconnected from AC power. As another example, the transducers **114** may be relatively smaller than transducers of the playback device **210** for example, so as to use relatively less power intensive to drive using the audio amplifiers **312h** during playback. Yet further, the software components **312c** may be configured to more aggressively enter power-saving modes as compared with other playback devices that are configured to operate using AC power. Other examples are possible as well.

[0117] FIGS. **3B** and **3C** are front and rear isometric views, respectively, of the portable playback device **310** configured in accordance with embodiments of the disclosed technology. FIGS. **3D** and **3E** are top and bottom views, respectively, of the portable playback device **310**.

[0118] Referring first to FIGS. **3A-3B**, the portable playback device **310** includes a housing **316** comprising an upper portion **316a**, a lower portion **316b** and an intermediate portion **316c** (e.g., a grille). A plurality of ports, holes or apertures **316d** in the upper portion **316a** allow sound to pass through to one or more microphones **315** positioned within the housing **316**. The one or more microphones **315** are configured to received sound via the apertures **316d** and produce electrical signals based on the received sound.

[0119] The user interface **313** includes a plurality of control surfaces (e.g., buttons, knobs, capacitive surfaces) on the top and rear of the portable playback device **310**. Referring to FIG. **3C**, the rear of the portable playback device includes a first button **313a** (power), a second button **313b** (a toggle for IEEE 802.11-and 802.15-compatible interfaces), and a third button **313c** (reset). Referring to FIG. **3D**, the top of the portable playback device **310** includes a first control surface **313d** (e.g., a previous control), a second control surface **313e** (e.g., a next control), and a third control surface **313f** (e.g., a play and/or pause control). A fourth control surface **313g** is configured to receive touch input corresponding to activation and deactivation of the one or microphones **315**.

[0120] The user interface **313** also includes indicators. A first indicator **313h** (e.g., one or more light emitting diodes (LEDs) or another suitable illuminator) can be configured to illuminate only when the one or more microphones **315** are activated. A second indicator **313i** (e.g., one or more LEDs) can be configured to remain solid during normal operation and to blink or otherwise change from solid to indicate a detection of voice activity. In some embodiments, the user interface **313** includes additional or fewer control surfaces and illuminators. In one embodiment, for example, the user interface **313** includes the first indicator **313e**, omitting the second indicator **313f**. Moreover, in certain embodiments, the NMD **320** comprises a playback device and a control device, and the user interface **313** comprises the user interface of the control device.

[0121] Referring to FIG. 3E, the lower portion **316** of the housing **316** includes first electrical contacts **318a**. FIG. 3F is an isometric view of a charging base **319**. When the portable playback device **310** is placed onto the charging base **319**, the first electrical contacts **318a** make contact with second electrical contracts **318b** of the charging base **319** completing a circuit within the power **312** of the portable playback device. The portable playback device **310** draws current from an AC outlet to operate and/or charge batteries of the power **312i**.

[0122] Referring to FIGS. 3A-3E together, the portable playback device **310** includes an NMD **320** is configured to receive voice commands from one or more adjacent users via the one or more microphones **315**. As described above with respect to FIG. 1B, the one or more microphones **315** can acquire, capture, or record sound in a vicinity (e.g., a region within **10m** or less of the NMD **320**) and transmit electrical signals corresponding to the recorded sound to the electronics **312**. The electronics **312** can process the electrical signals and can analyze the resulting audio data to determine a presence of one or more voice commands (e.g., one or more wake words). In some embodiments, for example, after detection of one or more suitable voice commands, the NMD **320** is configured to transmit a portion of the recorded audio data to another device and/or a remote server (e.g., one or more of the computing devices **106** of FIG. 1B) for further analysis.

[0123] The remote server can analyze the audio data, determine an appropriate action based on the voice command, and transmit a message to the NMD **320** to perform the appropriate action. For instance, a user may speak "Sonos, play Michael Jackson." The NMD **320** can, via the one or more microphones **315**, record the user's voice utterance, determine the presence of a voice command, and transmit the audio data having the voice command to a remote server (e.g., one or more of the remote computing devices **106** of FIG. 1B, one or more servers of a VAS and/or another suitable service). The remote server can analyze the audio data and determine an action corresponding to the command. The remote server can then transmit a command to the NMD **320** to perform the determined action (e.g., play back audio content related to Michael Jackson). The NMD **320** can receive the command and play back the audio content related to Michael Jackson from a media content source. As described above with respect to FIG. 1B, suitable content sources can include a device or storage communicatively coupled to the NMD **320** via a LAN (e.g., the network **104** of FIG. 1B), a remote server (e.g., one or more of the remote computing devices **106** of FIG. 1B), etc. In certain embodiments, however, the NMD **320** determines and/or performs one or more actions corresponding to the one or more voice commands without intervention or involvement of an external device, computer, or server.

[0124] FIGS. 4A-4D are schematic diagrams of a control device **430** (e.g., the control device **130** a of FIG. 1H, a smartphone, a tablet, a dedicated control device, an IoT device, and/or another suitable device) showing corresponding user interface displays in various states of operation. A first user interface display **431a** (FIG. 4A) includes a display name **433a** (i.e., "Rooms"). A selected group region **433b** displays audio content information (e.g., artist name, track name, album art) of audio content played back in the selected group and/or zone. Group regions **433c** and **433d** display corresponding group and/or zone name, and audio content information audio content played back or next in a playback queue of the respective group or zone. An audio content region **433e** includes information related to audio content in the selected group and/or zone (i.e., the group and/or zone

indicated in the selected group region **433b**). A lower display region **433f** is configured to receive touch input to display one or more other user interface displays. For example, if a user selects “Browse” in the lower display region **433f**, the control device **430** can be configured to output a second user interface display **431b** (FIG. 4B) comprising a plurality of music services **433g** (e.g., Spotify, Radio by Tunein, Apple Music, Pandora, Amazon, TV, local music, line-in) through which the user can browse and from which the user can select media content for play back via one or more playback devices (e.g., one of the playback devices **110** of FIG. 1A). Alternatively, if the user selects “My Sonos” in the lower display region **433f**, the control device **430** can be configured to output a third user interface display **431c** (FIG. 4C). A first media content region **433h** can include graphical representations (e.g., album art) corresponding to individual albums, stations, or playlists. A second media content region **433i** can include graphical representations (e.g., album art) corresponding to individual songs, tracks, or other media content. If the user selections a graphical representation **433j** (FIG. 4C), the control device **430** can be configured to begin play back of audio content corresponding to the graphical representation **433j** and output a fourth user interface display **431d** fourth user interface display **431d** includes an enlarged version of the graphical representation **433j**, media content information **433k** (e.g., track name, artist, album), transport controls **433m** (e.g., play, previous, next, pause, volume), and indication **433n** of the currently selected group and/or zone name.

[0125] FIG. 5 is a schematic diagram of a control device **530** (e.g., a laptop computer, a desktop computer). The control device **530** includes transducers **534**, a microphone **535**, and a camera **536**. A user interface **531** includes a transport control region **533a**, a playback status region **533b**, a playback zone region **533c**, a playback queue region **533d**, and a media content source region **533e**. The transport control region comprises one or more controls for controlling media playback including, for example, volume, previous, play/pause, next, repeat, shuffle, track position, crossfade, equalization, etc. The audio content source region **533e** includes a listing of one or more media content sources from which a user can select media items for play back and/or adding to a playback queue.

[0126] The playback zone region **533b** can include representations of playback zones within the media playback system **100** (FIGS. 1A and 1B). In some embodiments, the graphical representations of playback zones may be selectable to bring up additional selectable icons to manage or configure the playback zones in the media playback system, such as a creation of bonded zones, creation of zone groups, separation of zone groups, renaming of zone groups, etc. In the illustrated embodiment, a “group” icon is provided within each of the graphical representations of playback zones. The “group” icon provided within a graphical representation of a particular zone may be selectable to bring up options to select one or more other zones in the media playback system to be grouped with the particular zone. Once grouped, playback devices in the zones that have been grouped with the particular zone can be configured to play audio content in synchrony with the playback device(s) in the particular zone. Analogously, a “group” icon may be provided within a graphical representation of a zone group. In the illustrated embodiment, the “group” icon may be selectable to bring up options to deselect one or more zones in the zone group to be removed from the zone group. In some embodiments, the control device **530** includes other interactions and implementations for grouping and ungrouping zones via the user interface **531**. In certain embodiments, the representations of playback zones in the playback zone region **533b** can be dynamically updated as playback zone or zone group configurations are modified.

[0127] The playback status region **533c** includes graphical representations of audio content that is presently being played, previously played, or scheduled to play next in the selected playback zone or zone group. The selected playback zone or zone group may be visually distinguished on the user interface, such as within the playback zone region **533b** and/or the playback queue region **533d**. The graphical representations may include track title, artist name, album name, album year, track length, and other relevant information that may be useful for the user to know when controlling the

media playback system **100** via the user interface **531**.

[0128] The playback queue region **533d** includes graphical representations of audio content in a playback queue associated with the selected playback zone or zone group. In some embodiments, each playback zone or zone group may be associated with a playback queue containing information corresponding to zero or more audio items for playback by the playback zone or zone group. For instance, each audio item in the playback queue may comprise a uniform resource identifier (URI), a uniform resource locator (URL) or some other identifier that may be used by a playback device in the playback zone or zone group to find and/or retrieve the audio item from a local audio content source or a networked audio content source, possibly for playback by the playback device. In some embodiments, for example, a playlist can be added to a playback queue, in which information corresponding to each audio item in the playlist may be added to the playback queue. In some embodiments, audio items in a playback queue may be saved as a playlist. In certain embodiments, a playback queue may be empty, or populated but “not in use” when the playback zone or zone group is playing continuously streaming audio content, such as Internet radio that may continue to play until otherwise stopped, rather than discrete audio items that have playback durations. In some embodiments, a playback queue can include Internet radio and/or other streaming audio content items and be “in use” when the playback zone or zone group is playing those items.

[0129] When playback zones or zone groups are “grouped” or “ungrouped,” playback queues associated with the affected playback zones or zone groups may be cleared or re-associated. For example, if a first playback zone including a first playback queue is grouped with a second playback zone including a second playback queue, the established zone group may have an associated playback queue that is initially empty, that contains audio items from the first playback queue (such as if the second playback zone was added to the first playback zone), that contains audio items from the second playback queue (such as if the first playback zone was added to the second playback zone), or a combination of audio items from both the first and second playback queues. Subsequently, if the established zone group is ungrouped, the resulting first playback zone may be re-associated with the previous first playback queue, or be associated with a new playback queue that is empty or contains audio items from the playback queue associated with the established zone group before the established zone group was ungrouped. Similarly, the resulting second playback zone may be re-associated with the previous second playback queue, or be associated with a new playback queue that is empty, or contains audio items from the playback queue associated with the established zone group before the established zone group was ungrouped.

[0130] FIG. **6** is a message flow diagram illustrating data exchanges between devices of the media playback system **100** (FIGS. **1A-1M**).

[0131] At step **650a**, the media playback system **100** receives an indication of selected media content (e.g., one or more songs, albums, playlists, podcasts, videos, stations) via the control device **130a**. The selected media content can comprise, for example, media items stored locally on or more devices (e.g., the audio source **105** of FIG. **1C**) connected to the media playback system and/or media items stored on one or more media service servers (one or more of the remote computing devices **106** of FIG. **1B**). In response to receiving the indication of the selected media content, the control device **130a** transmits a message **651a** to the playback device **110a** (FIGS. **1A-1C**) to add the selected media content to a playback queue on the playback device **110a**.

[0132] At step **650b**, the playback device **110a** receives the message **651a** and adds the selected media content to the playback queue for play back.

[0133] At step **650c**, the control device **130a** receives input corresponding to a command to play back the selected media content. In response to receiving the input corresponding to the command to play back the selected media content, the control device **130a** transmits a message **651b** to the playback device **110a** causing the playback device **110a** to play back the selected media content. In response to receiving the message **651b**, the playback device **110a** transmits a message **651c** to the computing device **106a** requesting the selected media content. The computing device **106a**, in

response to receiving the message **651c**, transmits a message **651d** comprising data (e.g., audio data, video data, a URL, a URI) corresponding to the requested media content.

[0134] At step **650d**, the playback device **110a** receives the message **651d** with the data corresponding to the requested media content and plays back the associated media content.

[0135] At step **650e**, the playback device **110a** optionally causes one or more other devices to play back the selected media content. In one example, the playback device **110a** is one of a bonded zone of two or more players (FIG. **1M**). The playback device **110a** can receive the selected media content and transmit all or a portion of the media content to other devices in the bonded zone. In another example, the playback device **110a** is a coordinator of a group and is configured to transmit and receive timing information from one or more other devices in the group. The other one or more devices in the group can receive the selected media content from the computing device **106a**, and begin playback of the selected media content in response to a message from the playback device **110a** such that all of the devices in the group play back the selected media content in synchrony.

IV. Example Power Coordination

[0136] FIG. **7A** is a block diagram illustrating an example power coordinator system that includes a power coordinator **760**, multiple client programs **761**, and a syslib **762**, which may be implemented as in the software components **312c** of the portable playback device **310**. More particularly, the power coordinator **760** is implemented as a background process (i.e., a daemon) that launches during boot of an operating system of the software components **312c**. As explained in more detail below, the power coordinator coordinates between the multiple client programs **761**, and the syslib **762** to manage power coordination services, such as kernel suspend and power level.

[0137] Some or all of the programs executing on the playback device **310** may be configured as client programs **761** of the power coordinator **760** to facilitate participation in power coordination of the portable playback device **310**. By participating in power coordination, the client programs **761** influence various power-related operations, such as when the portable playback device **310** suspends and resumes, as well as the current power level of the portable playback device **310**. Client programs **761** may implement a power coordination library to facilitate interaction with the power coordinator **760** over the respective IPC mechanisms, possibly compiled as a shared object (e.g., a dynamically linked library).

[0138] As shown, the client programs **761** include client programs **761a**, **761b**, **761c**, and **761d**. By way of illustration, client program **761a** is a control plane program configured to control audio playback by the portable playback device **310**. Client program **761b** is a network manager configured to manage 802.11 connections via the network interface **312d**. Similarly, client program **761c** is a connection manager that, in operation, manages 802.15 connections via the network interface **312d**. Client program **761d** is an upgrade manager configured to manage upgrades to the software components **312c** of the portable playback device **310**. These client programs are shown by way of example, and other implementations may include additional or fewer client programs, as well as client programs that are responsible for different device functions.

[0139] In operation, the power coordinator **760** registers the multiple client programs **761** by establishing respective inter-process communication (IPC) mechanisms between the power coordinator **760** and the multiple client programs **761**. As shown by way of example in FIG. **7A**, the power coordinator **760** has established respective IPC mechanisms between the power coordinator **760** and the client programs **761a**, **761b**, **761c**, and **761d**. These IPC mechanisms allow the client programs **761a**, **761b**, **761c**, and **761d** and the power coordinator **760** to communicate with one another.

[0140] In some examples, a particular subset of the multiple client programs **761** may be considered critical. For instance, the client programs **761a** may be considered critical, as its responsible for audio playback. If the IPC mechanism(s) between the power coordinator **760** and the critical client programs **761** become disconnected, the power coordinator **760** might not know whether a kernel suspend is safe to perform (i.e., whether a suspend would interrupt functions of

the critical client program). In such circumstances the power coordinator **760** may prevent kernel suspend when the respective IPC mechanisms between the particular subset of client programs and the power coordinator background process are disconnected. When such disconnection occurs, the power coordinator **760** may attempt to reestablish the IPC mechanism(s) so that operation can continue normally.

[0141] The syslib **762** is a system library that provides the power coordinator **760** access to kernel functions using commands, function calls, protocols, and/or objects, which are collectively referred to as instructions in FIG. 7A. Example kernel functions include suspend and set power level, among others. To access such functions, the power coordinator **760** may make a function call to the appropriate function in the library. Certain client programs **761** (i.e., client program **761 a**) have access to the syslib **762** in addition to the power coordinator **760**. Such access facilitates communication with kernel drivers, which control hardware components of the portable playback device **310**. For example, as shown in FIG. 7A, client program **761a** has access to the syslib **762** to control kernel drivers corresponding to the audio pipeline (e.g., the audio processing components **312g** and audio amplifier(s) **312h**).

[0142] As further shown in FIG. 7A, the syslib **762** maintains a power event queue. In some examples, the power event queue may be implemented as a first-in-first-out data structure. Power events generated by the kernel are placed into the queue by the syslib **762**. In operation, the power coordinator **760** reads the power event queue for information on power events generated by the kernel. The power coordinator **760** may further distribute data indicating these events to the client programs **761**, which may modify their operation or take action based on this information.

[0143] One example power event is “resumed source event,” which is generated when the portable playback device **310** resumes from kernel suspend. Via this power event, the syslib **762** communicate the reason that the portable playback device **310** came out of kernel suspend. Various triggers to kernel resume are discussed in more detail below.

[0144] Other example power events relate to the battery. For instance, one example power event may be “charge state changed,” which is generated when the battery is connected to or disconnected from a charging source such as the charging base **319** (FIG. 3F). Another example power event is “battery fault,” which is generated when the battery is dead or when the battery has an issue such as overheating. A further example power event is a “battery voltage level (percentage) triggered,” which is generated when a certain battery voltage level is reached during charging or discharging. Yet another example power event is “battery voltage level critical,” which is generated when the battery voltage level reaches some pre-defined critical level (e.g., 1%). Other example power events may be generated in addition to these examples.

[0145] In an example, the power coordinator **760** receives, from the syslib **762**, charge state event data indicating a charge state of the battery, the charge state comprising one of: (a) the battery is connected to a charging source or (b) the battery is disconnected from the charging source. The power coordinator **760** may receive the charge state event data via a power event in the power event queue. The power coordinator **760** sends, via the established IPC mechanisms to the multiple client programs **761**, messages indicating the charge state of the battery. As noted above, the client programs **761** may modify their operation or take action based on such data. For instance, the client program **761a** may send a message to the power coordinator **760** indicating that the given client program is ready to suspend and a given resume time, wherein the given client program determines the given resume time based on the charge state of the battery.

[0146] FIG. 7B is a block diagram illustrating example inter-process communication between the power coordinator **760** and the client program **761a**. As shown in FIG. 7B, the example inter-process communication includes socket(s) **763** and shared memory **764**. The socket(s) **763** are a set of data communication endpoints for exchanging data (e.g., messages) between the power coordinator **760** and the client program **761a**. The shared memory **764** is a segment of memory accessible to both the power coordinator **760** and the client program **761a**. Since both the power

coordinator **760** and the client program **761a** can read and write to the shared memory **764**, the shared memory **764** can be used to communicate and share data. In some examples, the portable playback device **300** may implement a mutex, as shown. The mutex is set by the power coordinator **760** or the client program **761a** when accessing the shared memory, in an effort to prevent simultaneous access and accordant issues, such as race conditions.

[0147] As one example, the multiple client programs **761** may use the IPC mechanisms to communicate when to kernel suspend and resume. During kernel suspend, the main processor(s) of the portable playback device are suspended, and userspace programs like the power coordinator **760** and the client programs **761** are not executing, which reduces power consumption of the portable playback device **310**. However, since these programs are not executing, they cannot perform their intended functions.

[0148] Accordingly, the power coordinator **760** may implement a kernel suspend algorithm configured to determine when each client program is ready to kernel suspend. In example implementations, in an attempt to avoid interrupting operation of a client program **761**, the power coordinator refrains from initiating kernel suspend until every client program **761** is ready to suspend. In alternative implementations, certain client programs may be classified as a first type of client program that is permissible to interrupt, while other client programs are classified as a second type of client program for which interruption is prohibited. In operation, the power coordinator **760** may receive respective messages indicating that each client program **761** is ready to suspend. Based on these messages, the power coordinator **760** may determine that the portable playback device **310** may enter kernel suspend. Ultimately, to cause the portable playback device **310** to kernel suspend, the power coordinator **760** sends instructions to the operating system using the syslib **762** (e.g., by using a function call or command).

[0149] FIG. 7C is a timing diagram illustrating an example suspend algorithm that the power coordinator **760** may implement to determine when to kernel suspend. As shown in FIG. 7C, at time t_0 , the power coordinator **760** receives a message from client program **761a** indicating that client program **761a** is ready to suspend. The message also includes a time (t_7) when the client program **761a** must be resumed. Similarly, at times t_1 , t_2 , and t_3 , respectively, the power coordinator **760** receives messages from client programs **761c**, **761b**, and **761d** indicating that each client program is ready to suspend, along with the respective time that each client must be resumed. At time t_4 , the power coordinator **760** determines each client program of the multiple client programs is ready to suspend, and, based on this determination, sends instructions to the operating system to kernel suspend (e.g., via syslib **762**).

[0150] At time t_4 , the power coordinator **760** also sets a kernel suspend timeout trigger to the earliest resume time among the resume times indicated in the received messages from the multiple client programs, which in this example is time t_5 . Since the power coordinator **760** is not executing during kernel suspend, the kernel suspend timeout trigger is set via the syslib **762** on an auxiliary processor or secondary component that is executing during kernel suspend. Prior to time t_5 , other wake up triggers may cause the portable playback device **300** to resume. Further details regarding wake up triggers are discussed in more detail below.

[0151] FIG. 7D is a message flow diagram illustrating messages exchanged between the power coordinator **760**, the multiple client programs **761**, and the syslib **762** to facilitate the example suspend algorithm. In operation, each client program **761** may determine whether the respective client program is ready to kernel suspend. Generally, client programs **761** are ready to kernel suspend when they not in use (i.e., idle) and do not expect to be in use during the kernel suspend. For instance, the client program **761a** may determine that it is ready to kernel suspend when idle (i.e., not controlling the portable playback device **310** to play back audio) and when no commands to play back audio during the kernel suspend have been received. Since the portable playback device **310** uses significantly less power while in kernel suspend (as compared with not suspended), the multiple client programs **761** may repeatedly (e.g., periodically) determine whether

they are ready to suspend and communicate their respective status to the power coordinator **760** using their respective IPC mechanism in an attempt to reduce the overall power use of the portable playback device **310**.

[0152] At **771a**, the power coordinator **760** receives messages from the client program(s) **761** indicating that the respective client program(s) **761a-d** are ready to suspend. These messages may also include the respective times that each client program must be resumed. Based on these messages, at **772a**, the power coordinator **760** determines that each client program **761a-d** of the multiple client programs **761** is ready to suspend.

[0153] In this example, when the power coordinator **760** determines that each client program(s) **761a-d** is ready to suspend, at **771b**, the power coordinator **760** initiates a suspend by sending a suspend imminent event to the multiple client program(s) **761**. The suspend imminent event notifies the client programs **761a-d** of the forthcoming suspend and provides another opportunity for the client programs **761** to set a new resume time or power level.

[0154] After sending out the suspend imminent event, the power coordinator **760** waits a pre-determined period of time (e.g., ten seconds) for messages from the client programs requesting a new resume time or power level. During this waiting period, client programs **761** can raise or lower their respective power level requirements, which does not terminate the suspend as long as each client program remains in the ready to suspend state at the end of the suspend imminent waiting period. Further details regard raising and lowering power levels are discussed below.

[0155] After waiting, if the multiple client program(s) **761** remain in the ready to suspend state, the power coordinator **760** confirms the suspend at **772b**. At this same time, the power coordinator **760** also determines the earliest resume time among the resume times indicated in the received messages from the client program(s) **761**. For instance, in the FIG. 7C example, the earliest time was time **t5**. At **771c**, the power coordinator **760** sends instructions to the syslib **762** to initiate a kernel suspend and also to set a kernel suspend timeout trigger to the determined earliest resume time among the resume times indicated in the received messages from the client program(s) **761**.

[0156] At **773a**, the syslib **762** receives the instructions and causes the portable playback device **310** to kernel suspend. As noted above, during suspend, user spaces programs like the power coordinator **760** and the client programs **761** are not executing and the main processors are in a suspended state. The portable playback device **310** remains in suspend until a trigger to kernel resume is detected at **773b**. The kernel suspend timeout trigger will resume the portable playback device **310** unless a different trigger to kernel resume is detected prior to the kernel suspend timeout trigger. The syslib **762** adds a suspend resumed event indicating the source of the kernel resume trigger to the power event queue to facilitate distributing this source among the userspace programs.

[0157] At **773c**, the kernel is resumed. Once the kernel and drivers are ready, userspace programs like the power coordinator **760** and the client program(s) **761** are resumed as well. At **772c**, when the power coordinator **760** is executing, the power coordinator **760** reads the power event queue, which includes the suspend resumed event indicating the source of the kernel resume trigger.

[0158] At **771d**, the power coordinator **760** distributes data indicating the source of the kernel resume trigger to the multiple client programs **761**. This allows the client programs to modify their operation based on the source of the kernel resume trigger. For instance, if the source of the kernel resume trigger is a wake-on-BLE (Bluetooth LE) trigger, the client program **761a** may change the playback source to streaming via Bluetooth LE. Other examples are possible as well.

[0159] FIG. 7E is a timing diagram illustrating an example where a kernel suspend is attempted but aborted. As shown, the timing diagram of FIG. 7E begins with a suspend imminent event **771b** send to the multiple clients **761**. The power coordinator **760** may send the suspend imminent event **771b** when the multiple client programs **761** have indicated respective ready to suspend states or when the user input indicates a request to suspend (e.g., via a short press of the power button **313a** (FIG. 3C)).

[0160] After waiting a pre-determined period of time (here, six seconds) for messages from the client programs requesting a new resume time or power level, the power coordinator **760** performs a suspend confirm **772b**, which is an attempt to confirm the suspend. Here, the power coordinator **760** is unable to confirm the suspend. One reason that the power coordinator **760** might be unable to confirm the suspend is that the earliest mustWakeTime is too close to the current time such that the portable playback device **310** would be unable to resume by the earliest mustWakeTime (or the suspend would be inefficient). Other reasons that the power coordinator **760** might be unable to confirm the suspend include kernel errors, driver faults, and the like. Since the power coordinator **760** is unable to confirm the suspend, the power coordinator **760** sends the suspend abort event **772d**, which notifies the multiple client programs **761** that the suspend has been aborted.

[0161] FIG. 7F is a timing diagram illustrating an example where a kernel suspend is completed. As shown, the timing diagram of FIG. 7E begins with a suspend imminent event **771b** send to the multiple clients **761**. After waiting a pre-determined period of time, the power coordinator **760** performs a suspend confirm **772b**, which is an attempt to confirm the suspend. Here, the power coordinator **760** is able to confirm the suspend. According, the power coordinator **760** sends the suspend committed event **772e**, which sends instructions to the syslib **762** to kernel suspend. A short time later, the userspace programs (e.g., the power coordinator **760** and the multiple client programs **761**) suspend at the userspace suspend **773f**. At this time, the kernel and the userspace programs are suspended.

[0162] The kernel resume **773c** occurs in response to one of the triggers to kernel resume, such as the kernel suspend timeout trigger, among others. After the kernel is resumed, the userspace resume **773d** occurs less than a second later. The power coordinator resume **773e** occurs a short time later as the various userspace programs are resumed. Once the power coordinator **760** is executing, the system resume event **773d** involves the power coordinator **760** reading the kernel resume event from the power event queue populated by the syslib **762** and distributing data indicating the source of the kernel resume trigger to the multiple client programs **761**.

[0163] Within examples, the power coordinator **760** may also set the power level of the portable playback device **310**. The portable playback device **310** may support multiple power levels, referred to herein as a first power level, a second power level, a third power level, and so on. As noted above, a “power level” refers to a specific configuration of settings that when set, results in a given hardware performance for a particular power cost.

[0164] Various hardware settings influence power usage. Processor settings that may contribute to power level include processor frequency, number of processor cores enabled/disabled, and processor voltage, among others. Other settings that contribute to power mode include enabling/disabling components, such as the network interface **312d**, audio processing components **312g**, audio amplifiers **312h**, and/or the other components **312j**. Certain of these components may support low power modes as well. These settings may be configured by the power coordinator **860** using the syslib **862**, which provides access to the kernel drivers for the corresponding components.

[0165] In some implementations, the power coordinator **760** implements a power level algorithm to set the power level of the portable playback device **310**. In an example, the power coordinator **760** receives, via the established IPC mechanisms from the multiple client programs, messages indicating respective power level requirements of the multiple client programs **761**. For instance, the client program **761a** may send a message indicating that it requires the first power level, while the client programs **761b-d** send respective messages indicating that they require relatively lower power levels, such as the second or third power levels.

[0166] Based on the received messages from the multiple client programs **761**, the power coordinator **760** determines a particular power level from among the power levels indicated by each client program **761a-d**. This particular power level is the highest power level among the respective power level requirements of the multiple client programs **761**, so that each client program **761a-d** has available sufficient resources for performing its corresponding functions. For instance,

continuing the example above, the power coordinator **760** may determine that the first power level is appropriate, as the client program **761a** requires this highest power level.

[0167] After determining the particular power level, the power coordinator **760** send instructions to the operating system to operate at the particular power level. Sending instructions to the operating system may involve making a specific combination of function calls to the syslib **762** to configure the configuration of settings that when set, results in the components of the portable playback device **310** operating at the particular power level. For instance, the power coordinator may make a first function call to set processor frequency, a second function call to enable certain processor cores, a third function call to enable the network interface(s) **312g**, and so on.

[0168] FIG. 7G is a table showing example power levels and their corresponding settings. These examples are for purposes of illustration-commercial implementations may use different combination of settings, as well as different power levels. Other examples are possible as well.

V. Example Kernel Resume Triggers

[0169] FIG. 8A is a block diagram showing an example system **880**, which illustrates example architecture of certain components of the portable playback device **310**. As discussed above, the power coordinator **760** determines when to kernel suspend. However, since the kernel and userspace programs, such as the power coordinator **760**, are not executing during kernel suspend, kernel resume triggers must be external to the kernel and userspace programs. The architecture of system **880** facilitates kernel resume based on a plurality of kernel resume triggers from different components.

[0170] System **880** includes a system-on-chip (SoC) **881**, which is an implementation of the processors **312a** of the portable playback device **310** (FIG. 3A). SoC **881** includes main processor(s) **881** and auxiliary processor **882**. In this example, the main processor(s) **881a** and auxiliary processor **881b** are implemented as separate cores on the SoC **881**. The kernel executes on the SoC **881**, as well as various userspace programs executing on top of the kernel.

[0171] To facilitate various power levels, the main processor(s) **881a**, as well as the auxiliary processor **881b**, are individually power-and clock-gated. In various examples, the kernel can distribute the workload of the portable playback device **310** to the available cores. During kernel suspend, the main processors **881** are power gated to reduce power usage but the auxiliary processor **881b** is kept active to receive interrupts that trigger kernel resume. The main processors **881a** are resumed by the auxiliary processor **881b**.

[0172] System **880** also includes a programmable system-on-chip (PSoC) **882**, which is separately programmable from the SoC **881**. PSoC **882** includes capacitive touch (CapTouch) **882a** and Bluetooth Low Energy (BLE) **882b**. PSoC **882** may include other components as well.

[0173] CapTouch **882a** includes a driver and corresponding circuitry configured to receive input data from the capacitive touch portions of the user interface **313**, such as the first control surface **313d**, the second control surface **313e**, the third control surface **313f**, and fourth control surface **313g** (FIG. 3D). The CapTouch **882a** driver interprets this data and generates corresponding commands, which are shared with the SoC **881** and ultimately the client programs **761** to control various functions.

[0174] The BLE **882b** includes a driver and corresponding circuitry configured to manage Bluetooth Low Energy connections from smartphones, tablets, and other electronic devices. In operation, the portable playback device may play back audio streams received via such BLE connections. In some examples, the portable playback device may also receive data representing transport controls, such as play, pause, and skip, via a BLE connection.

[0175] During kernel suspend, the PSoC **882** remains active to generate kernel resume triggers (e.g., interrupts). For example, in some implementations, the CapTouch **882a** generates an interrupt when touch input is received via the user interface **313**, which is sent to the auxiliary processor **881b** to facilitate a wake-on-touch kernel resume event. As another example, when a BLE connection is made, the BLE **882b** generates an interrupt, which is likewise sent to the auxiliary

processor **881b** to facilitate a wake-on-BLE kernel resume event.

[0176] Some implementations of the PSoC **882** include one or more sleep modes configured to resume the SoC **881**. For instance, in a first sleep mode, the PSoC **882** is configured such that the capacitive touch portions of the user interface **313** are ganged together to form a single proximity detector. In this first sleep mode, capacitive touch input to any control surface of the user interface **313** triggers an interrupt to resume the SoC **881**. In a second sleep mode, the PSoC **882** is configured such that only one capacitive touch portion of the user interface **313** is enabled. In this second sleep mode, capacitive touch input to the enabled control surface of the user interface **313** triggers an interrupt to resume the SoC **881**. Alternatively, during kernel suspend, the PSoC remains fully active and any touch input triggers interrupt to resume the SoC **881**.

[0177] System **880** further includes a power management microcontroller (uC) **883**, which is part of power **312i** (FIG. 3A). The power management uC **883** includes a power manager **884**, which is a firmware that manages the power source (AC or battery power), delivering power via the power rails (e.g., 3V, 5V, 12V, etc.), and charging via the charger **885**, among other functions. The charger **885** includes one or more of a charging base **319** or an interface that can deliver current to the portable playback device **310**, such as a universal serial bus port (e.g., USB Type C or micro-USB). The power manager **884** also generates battery related events, such as “charge state changed,” “battery fault,” “battery voltage level (percentage) triggered,” and “battery voltage level critical.”

[0178] The power management uC **883** is also connected to a button **886**, which may be implemented as the power button **313a** (FIG. 3C). To support wake-on-button, the power manager **884** is configured to detect a button press (e.g., a short button press). When a button press is detected, the power manager **884** sends a kernel resume event (e.g., an interrupt) to the auxiliary processor **881b** of the SoC **881**, which causes kernel resume as discussed above.

[0179] System **880** also includes a Wi-Fi (802.11-compatible) chipset **887** that includes a wireless network interface. The wireless network interface supports Wi-Fi connections over an 802.11-compatible standard such as 802.11 ac and/or 802.11 a/b/g/n. The Wi-Fi **887** generates wake-on-wireless events (e.g., interrupts) when a wake-on-wireless magic packet is received via the wireless network interface. The power manager **884** receives these events and sends a corresponding event to the auxiliary processor **881b** to wake up the main processors **881a** and resume the kernel.

[0180] System **880** also includes a real time clock (RTC) **888**. The RTC **888** manages kernel resume triggers based on clock time, such as alarms and the kernel resume timeout trigger. In an example, the RTC **888** is implemented in the PSoC **882**, as the PSoC **882** is already awake during kernel suspend to support kernel resume triggers from CapTouch **882a** and/or BLE **882b**. Alternatively, the RTC **888** may be implemented in a different integrated circuit, such as the power management uC or a dedicated RTC IC.

[0181] In an example, alarms are set via the client program **761a**. The client program **761a** in turn sets the alarm in the RTC **888** via instructions to syslib **762** (FIG. 7A). At the time of the alarm, the RTC generates a kernel resume event (e.g., an interrupt). Similarly, the power coordinator **760** sets a kernel suspend timeout trigger in the RTC **888** via the syslib **762**. During a kernel suspend, if no other kernel resume trigger occurs, the portable playback device **310** is resumed via the kernel suspend timeout trigger generated by the RTC **888**.

[0182] FIG. 8B is a block diagram showing an example hierarchy illustrating how the power coordinator **760** is notified of the source of kernel resume. On the lowest level of the hierarchy are example kernel resume sources, including a wake on-button **886**, wake on charger **885**, wake on battery (e.g., critical or fault), wake on RTC **888** alarm, wake on BLE **882b** and wake on wireless. On the second level of the hierarchy are the kernel drivers that handle interrupts from each of the corresponding kernel resume sources and responsively wake the kernel. When the kernel is resumed, the syslib **762** adds a “resumed source event” power event to the power event queue. The “resumed source event” power event indicates the source of the kernel resume trigger. Once the power coordinator **760** is resumed, the power coordinator reads the power event queue to determine

the source of the kernel resume. As noted above, the power coordinator **760** may notify the client programs **761** of the source of the kernel resume via the IPC mechanisms.

[0183] FIG. **9A** is a flow diagram illustrating an example method **991** to facilitate wake-on-battery. The method **991** begins at **991a** with the power manager **884** executing on the power management uC **883**. At this time, the kernel is suspended (which places the main processors **881** a in a suspended (e.g., power gated) state).

[0184] At **991b**, the power manager **884** determines whether a battery fault has occurred. Battery faults may occur for various reasons, such as battery dead or overheating. The power management uC **883** includes various sensors to detect a battery fault, such as voltage, current, and/or temperature sensors, among others. If a battery fault has occurred, the power manager **884** sends an interrupt to wake the main processor(s) **881a** at **991e**. If no battery fault has occurred, the method **991** continues to **991c**.

[0185] At **991c**, the power manager **884** determines whether the battery level of the batteries is at critical level (e.g., 1%). The power management uC **883** includes sensors to detect battery level, such as voltage and/or current sensors. If the batteries are at critical level, the power manager **884** sends an interrupt to wake the main processor(s) **881a** at **991e**. If no battery fault has occurred, the method **991** continues to **991d**.

[0186] At **991d**, the power manager **884** determines whether the batteries have begun charging. The power management uC **883** includes one or more sensors to monitor charging status. If the batteries have started charging, the power manager **884** sends an interrupt to wake the main processor(s) **881a** at **991e**. If the charging status has not changed, the method **991** continues back to **991a**.

[0187] After the main processor(s) **881** are woken up at **991a**, the kernel resumes. At **991f**, Syslib **762** executing on the main processor(s) **881** sends a resume source power event indicating the source of the resume (e.g., wake on battery fault/critical battery level/begin charging) to the power event queue. Then, when the power coordinator **760** resumes, the power coordinator **760** can read the power event queue to determine the source of the kernel resume, as discussed above.

[0188] FIG. **9B** is a flow diagram illustrating an example method **992** to facilitate wake on button press. The method **992** begins at **992a** with the power manager **884** executing on the power management uC **883**. At this time, the kernel is suspended (which places the main processors **881a** in a suspended (e.g., power gated) state).

[0189] At **992b**, the power manager **884** determines whether the button **886** has been pressed (e.g., the power button **313a** (FIG. **3C**)). If the button **886** has been pressed in the programmed manner, the power manager **884** sends an interrupt to wake the main processor(s) **881a** at **992c**. If not, the method **992** continues back to **992a**.

[0190] In an example, the button **886** may be connected to the power management uC **883** via a circuit. Actuation of the button **886** may close or open the circuit. The power manager **884** monitors the status of this circuit to detect button presses. Further, in some implementations, the power manager **884** may differentiate between button presses of different lengths. For instance, a short press may be less than one second while a long press is greater than one second. In an example, the short press is programmed to wake-up while the long press is programmed to toggle power on or off.

[0191] After the main processor(s) **881** are woken up at **992c**, the kernel resumes. At **992d**, Syslib **762** executing on the main processor(s) **881** sends a resume source power event indicating the source of the resume (i.e., button press) to the power event queue. Then, when the power coordinator **760** resumes, the power coordinator **760** can read the power event queue to determine the source of the kernel resume, as discussed above.

[0192] FIG. **9C** is a flow diagram illustrating an example method **993** to facilitate wake on capacitive touch. The method **993** begins at **993a** with the CapTouch **882a** driver executing on the PSoC **882a**. At this time, the kernel is suspended (which places the main processors **881a** in a suspended (e.g., power gated) state).

[0193] At **993b**, the CapTouch **882a** driver determines whether a capacitive touch input has been received. As described above, during kernel suspend, the CapTouch **882a** may operate in various modes, such as a first sleep mode where the capacitive touch portions of the user interface **313** are ganged together to form a single proximity detector or a second sleep mode where only one capacitive touch portion of the user interface **313** is enabled. Alternatively, during kernel suspend, the PSOC remains fully active to detect capacitive touch input via the control surfaces of the user interface **313**. If capacitive touch input has been received, the CapTouch **882a** driver sends an interrupt to wake the main processor(s) **881a** at **993c**. If not, the method **993** continues back to **993a**.

[0194] After the main processor(s) **881** are woken up at **993c**, the kernel resumes. At **993d**, Syslib **762** executing on the main processor(s) **881** sends a resume source power event indicating the source of the resume (i.e., capacitive touch) to the power event queue. Then, when the power coordinator **760** resumes, the power coordinator **760** can read the power event queue to determine the source of the kernel resume, as discussed above.

[0195] FIG. **9D** is a flow diagram illustrating an example method **994** to facilitate wake on real time clock. The method **994** begins at **994a** with the RTC **888** driver executing. At this time, the kernel is suspended (which places the main processors **881a** in a suspended (e.g., power gated) state).

[0196] At **994b**, the RTC **888** driver determines whether an RTC interrupt has been generated. As indicated above, the RTC **888** driver may generate an RTC interrupt when at certain times. For instance, the RTC **888** driver may generate an RTC interrupt at expiration of the kernel suspend timeout trigger, which is set by the power coordinator **760** via the syslib **762**. As another example, the RTC **888** driver may generate an RTC interrupt at a time corresponding to an alarm, which may be set via a client program, such as client program **761a**. If the RTC interrupt has been generated, the RTC **888** driver sends an interrupt to wake the main processor(s) **881a** at **994c**. If not, the method **994** continues back to **994a**.

[0197] After the main processor(s) **881** are woken up at **994c**, the kernel resumes. At **994d**, Syslib **762** executing on the main processor(s) **881** sends a resume source power event indicating the source of the resume (i.e., RTC interrupt) to the power event queue. Then, when the power coordinator **760** resumes, the power coordinator **760** can read the power event queue to determine the source of the kernel resume, as discussed above.

[0198] FIG. **9E** is a flow diagram illustrating an example method **995** to facilitate wake on Bluetooth Low Energy. The method **995** begins at **995a** with the BLE **882b** driver executing on the PSoC **882**. At this time, the kernel is suspended (which places the main processors **881a** in a suspended (e.g., power gated) state).

[0199] At **995b**, the BLE **882b** driver determines whether a BLE connection request has been received. If the BLE connection request has not been received, the method **995** continues back to **995a**. If the BLE connection request has been received, at **995c**, the BLE **882b** driver determines whether the BLE connection has been successful. If the BLE connection request was not successful, the method **995** continues back to **995a**. If the BLE connection was successful, the BLE **882b** driver sends an interrupt to wake the main processor(s) **881a** at **995d**.

[0200] After the main processor(s) **881** are woken up at **995d**, the kernel resumes. At **995e**, Syslib **762** executing on the main processor(s) **881** sends a resume source power event indicating the source of the resume (i.e., wake on BLE) to the power event queue. Then, when the power coordinator **760** resumes, the power coordinator **760** can read the power event queue to determine the source of the kernel resume, as discussed above.

[0201] FIG. **9F** is a flow diagram illustrating an example method **996** to facilitate wake on wireless. The method **996** begins at **996a** with the Wi-Fi chipset **887** driver executing on the Wi-Fi chipset **887**. At this time, the kernel is suspended (which places the main processors **881a** in a suspended (e.g., power gated) state).

[0202] At **996b**, the Wi-Fi chipset **887** driver determines whether a wake-on-wireless (WoW) magic packet has been received. If the WoW magic packet has been received, the Wi-Fi chipset **887** driver sends an interrupt to wake the main processor(s) **881a** at **996c**. If not, the method **996** continues back to **996a**.

[0203] After the main processor(s) **881** are woken up at **996c**, the kernel resumes. At **996d**, Syslib **762** executing on the main processor(s) **881** sends a resume source power event indicating the source of the resume (i.e., WoW) to the power event queue. Then, when the power coordinator **760** resumes, the power coordinator **760** can read the power event queue to determine the source of the kernel resume, as discussed above.

[0204] FIG. **10** is a table illustrating example power modes of the portable playback device **310** as well as example status' of the various components of the portable playback device **310** in each power mode. These power modes are provided for purposes of illustration. Various embodiments may include additional or fewer power modes, different power modes, or different status of the various components of the portable playback device **310** in each power mode.

[0205] In the “Dead Battery/Shipping Power” mode, the power manager **884** has switched a battery discharged protection switch to open, which fully disconnects the battery from the electronics **312** of the portable playback device **310**. As shown in FIG. **10**, the components of the portable playback device **310** are all off. Once the charger **885** is connected, the power manager **884** will begin executing and transition the portable playback device **310** to “Off” mode.

[0206] In the “Off” mode, the portable playback device **310** is awaiting a button **886** press (e.g., a long press) to initiate BOOT of the kernel to transition to “Standby” mode. As shown in FIG. **10**, the power management uC **883** is active in “Off” mode, which allows the power manager **884** to execute. Other components are disabled.

[0207] In the “Standby” mode, the portable playback device **310** has an active wireless connection (to receive network communications from the control device **130** and/or the playback device(s) **110**) but no audio is playing. Certain components are active to support kernel resume triggers. If “Standby” mode persists for a pre-determined period of time (e.g., 20-30 minutes), the power manager **884** transitions the portable playback device into “Deep Sleep” mode to conserve battery.

[0208] In the “Deep Sleep” mode, the portable playback device **310** is waiting a button **886** press (e.g., a short press) to transition back to Standby mode. If “Deep Sleep” mode persists for a pre-determined period of time (e.g., 2-4 hours), the power manager **884** transitions the portable playback device into “Off” mode to conserve battery.

[0209] In “Playback” mode, the portable playback device **310** is playing audio. Most components are active to facilitate playback and associated operations, such as network-based audio streaming and control. As shown, the voice components (i.e., the NMD **320**) are in low power state to conserve battery.

[0210] In “Charging” mode, the portable playback device **310** is connected to charger **885**. Generally, components are active in “Charging” mode as power usage does not necessarily need to be minimized. If the charger **885** is disconnected, the power manager **885** transitions the portable playback device **310** to “Standby” mode.

[0211] In some implementations, in “Charging” mode, the portable playback device **310** operations in a zone of the media playback system **100**. This zone is programmatically assigned using the control device **130** and is generally based on the location of the charging base **319**. For instance, if the charging base **319** is located in the kitchen, the portable playback device **310** is generally assigned to the Kitchen zone.

VI. Example Power Coordination Techniques

[0212] FIG. **11** is a flow diagram showing an example method **1100** to facilitate power coordination in a portable playback device, such as the portable playback device **310** (FIGS. 3A-3F).

[0213] At block **1102**, the method **1100** includes launching a power coordinator. For example, the

portable playback device **310** may launch the power coordinator **760** (FIG. 7A), which is a power coordinator background process having multiple client programs **761a-d**. In some implementations, the kernel of the portable playback device **310** may launch (e.g., execute) the power coordinator **760** during the boot process or after a kernel resume following a kernel suspend, among other instances.

[0214] As described above, the multiple client programs **761a-d** may include programs configured to manage various functionality of the portable playback device **310**. In an example, the client programs **761a-d** include comprise a control program that configures the playback device to play back audio via the one or more speakers and the one or more amplifiers, a network manager program, an update manager program, and a Bluetooth manager program. Other examples are possible as well.

[0215] At block **1104**, the method **1100** includes establishing inter-process communication (IPC) mechanisms between the multiple client programs and the power coordinator. For instance, as shown in FIG. 7A, the power coordinator **760** may establish respective IPC mechanisms between the power coordinator **760** the multiple client programs **761a-d**. In example implementations, an IPC mechanism may include one or more sockets **763** and/or shared memory **764** (FIG. 7B). Other types of IPC mechanisms may be suitable as well.

[0216] At block **1106**, the method **1100** includes performing a kernel suspend. During a kernel suspend, the portable playback device **310** may disable one or more processors and/or decrease frequency of certain processors. For instance, the kernel may be configured to execute on the main processor(s) **881a** of the SoC **881** (FIG. 8A). To kernel suspend, the kernel may send instructions to the power manager **884** to disable the main processor(s) **881a**.

[0217] Within example implementations, the power manager **760** may determine when to kernel suspend based on input from the multiple client processes **761**. For instance, the power manager **760** may implement the example kernel suspend technique. (FIGS. 7C and 7D). In alternative implementations, the power manager **760** may implement another suitable algorithm.

[0218] As noted above, the example kernel suspend technique may involve the power manager **760** receiving, via the established IPC mechanisms from the multiple client programs **761**, messages indicating (a) that the respective client program **761a-d** is ready to suspend and (b) a respective resume time. The kernel suspend technique further involves determining, based on the received messages from the multiple client programs **761**, that each client program **761a-d** of the multiple client programs **761** is ready to suspend. Based on determining that each client program **761a-d** of the multiple client programs **761** is ready to suspend, the power manager **760** sends, via the established IPC mechanisms to the multiple client programs **761**, respective messages indicating that kernel suspend is imminent and then waits a pre-determined time period from sending the respective messages indicating that kernel suspend is imminent. When each client program **761a-d** of the multiple client programs **761** is ready to suspend after the pre-determined time period, the power manager **760** sends the instructions to the operating system to kernel suspend (e.g., via the syslib **762**). The power manager **760** may also set a kernel suspend timeout trigger to the earliest resume time among the resume times indicated in the received messages from the multiple client programs.

[0219] In further examples, the power coordinator **760** may estimate a resume time based on a usage history of the playback device. Such an estimation may prevent an unnecessary or inefficient short suspend, as the power coordinator **760** may determine not to kernel suspend (even if all clients are ready to suspend) if the estimated system resume time is not at least a threshold period of time in advance of the current time. Further the portable playback device **310** may alter the pre-determined period of times for waiting after the suspend imminent, or to transition between power modes, based on the usage history. For instance, if the portable playback device is frequency woken up via button press after going to “Deep Sleep” (FIG. 10), the portable playback device **310** may lengthen the pre-determined period of time to wait in “Standby” before going to “Deep Sleep.”

Other examples are possible as well.

[0220] Each client program **761a-d** may determine that its ready to suspend based on its own functions. For instance, a network manager client program or a Bluetooth manager client program may determine that the IEEE 802.11-compatible or 802.15-compatible interface is idle. Such a program may also determine (e.g., estimate) a time that the corresponding interface is expected to receive transmissions (e.g., packets) based on scheduled operations, historical usage data, or other factors. Based on such determinations, the client program may send the power manager **760**, one or more messages indicating (i) that the client program is ready to suspend and (ii) a resume time that is a pre-determined period of time before the estimated time that the corresponding interface is expected to receive transmissions.

[0221] As another example, an update manager client program may determine that no updates available from one or more servers (e.g., the computing devices **106** (FIG. **1B**)) and/or updates are scheduled at a certain update time. Based on this determination, the update client manager may send, to the power manager **760** via the established IPC mechanism, a message indicating (i) that the update manager program is ready to suspend and (ii) a resume time that is a particular period of time from a current time. The particular period may be a pre-determined period of time or a determined time based on the scheduled update time.

[0222] In another example, the control client program that configures the portable playback device **310** to play back audio may determine that no audio playback is ongoing or anticipated. The control client program may anticipate playback based on detecting user proximity (e.g., via a capacitive touch user interface) (FIG. **3D**), based on scheduled playback (e.g., zone scenes or alarms), or based on incoming control instructions from the control device **120**, among other examples. during audio playback, the control client program foregoes sending a message indicating that the messages indicating the control plane program is ready to suspend, as kernel suspend would interrupt the audio playback.

[0223] At block **1108**, the method **1100** includes detecting a particular trigger to kernel resume. Example triggers to kernel resume include the kernel suspend timeout trigger (e.g., from the RTC **888**), wake-on-button (e.g., from the button **886** via the power manager uC **883**), wake-on-battery (e.g., via the power manager uC **883**), wake-on-BLE or capacitive touch (e.g., form the CapTouch **882a** or BLE **882b** of the PSoC **882**), and wake-on-wireless (e.g., via the Wi-Fi chipset **887**) (FIGS. **8A** and **8B**). Example techniques to detect such triggers are described in FIGS. **9A-9F**, as well as throughout the disclosure. Additional or alternative triggers may be implemented as well.

[0224] At block **1110**, the method **1100** includes performing a kernel resume. For instance, the auxiliary processor **881b** of the SoC **881** may receive an interrupt from a kernel resume source (FIGS. **8A** and **8B**) and resume the main processor(s) **881a** and the kernel. Other examples are possible as well.

VII. Example Power Management Techniques

[0225] FIG. **12** is a flow diagram showing an example method **1200** to facilitate power management in a portable playback device, such as the portable playback device **310** (FIGS. **3A-3F**).

[0226] At block **1202**, the method **1200** involves monitoring for resume triggers during kernel suspend. As noted above, during kernel suspend, one or more main processors are in suspend mode (e.g., power-gated) and the kernel and userspace programs are not executing. Accordingly, one or more secondary controllers of the portable playback device **310** may monitor for appropriate conditions corresponding to multiple resume triggers.

[0227] Example resume triggers include wake-on-button (e.g., a power button, to facilitate the user requesting a resume), wake-on-battery (e.g., based on a fault or change in status), wake-on-Bluetooth, wake-on-wireless, and/or wake-on-touch. Example secondary controllers include the power management uC **883**, the PSoC **882**, and the Wi-Fi chipset **887** (FIG. **8A**), among other examples. Such secondary controllers may monitor for such resume triggers using the example

techniques illustrated in FIGS. 9A, 9B, 9C, 9D, 9E, and 9F, among other examples.

[0228] For instance, during kernel suspend, the power manager **884** may monitor, via one or more sensors of the power management uC **883**, the battery for conditions corresponding to respective wake-on-battery triggers (FIG. 9A). Example wake-on-battery triggers include a battery fault, a critical battery level, and battery charging initiated. In further examples, the power manager **884** may monitor, via the one or more sensors of the power management uC **883**, a circuit for a signal corresponding to a wake-on-button trigger (FIG. 9B). Such a signal may correspond to a press of the power button **313a** (FIG. 3C).

[0229] As another example, the PSoC **882** may monitor for various resume triggers. For example, the CapTouch **882a** driver may monitor a capacitive-touch user interface (e.g., control surfaces **313d-313g** in FIG. 3D) for data indicating a wake-on-touch trigger (FIG. 9C). As another example, the BLE **882b** driver may monitor, via an 802.15-compatible Bluetooth Low Energy (BLE) interface, for a wake-on-BLE trigger (FIG. 9E).

[0230] Another example involves the RTC **888** monitoring a clock of the RTC chipset for the kernel suspend triggers based on time (FIG. 9D). Example kernel suspend triggers based on time include the kernel suspend timeout trigger, alarms, and pre-programmed playback such as zone scenes. For instance, before kernel suspend of the kernel, the RTC **888** may receive, via the kernel, instructions from the power coordinator **760** to set a kernel suspend timeout trigger for a particular time and responsively set the kernel suspend timeout trigger.

[0231] In a further example, the Wi-Fi chipset **887** may monitor for a wake-on-wireless trigger (FIG. 9F). Monitoring for a wake-on-wireless trigger involve determining whether a wake-on-wireless magic packet has been received via the Wi-Fi chipset **887**. Other examples are possible as well.

[0232] At block **1204**, the method **1200** involves detecting a particular resume trigger. For instance, a given secondary controller may detect one of the example resume triggers, such as wake-on-button, wake-on-battery, wake-on-Bluetooth, wake-on-wireless, or wake-on-touch, among other examples. The example secondary controllers may monitor for such resume triggers using the example techniques illustrated in FIGS. 9A, 9B, 9C, 9D, 9E, and 9F, among other examples.

[0233] At block **1206**, the method **1200** involves sending an interrupt to resume the kernel. For example, the secondary controller may send an interrupt to an auxiliary processor **881b**, which resumes the main processors **881a** and the kernel in response to receiving the interrupt (FIG. 8A). Other examples are possible as well.

[0234] At block **1208**, the method **1200** involves distributing resume source to userspace. Example userspace programs include the power coordinator **760** and the multiple client programs **761** (FIG. 7A). Other userspace programs are possible as well.

[0235] In an example, the syslib **762** of the kernel adds a kernel resume event to a power event queue (FIG. 7A). This kernel resume event indicates the source of the kernel resume. After resuming from suspend, the power coordinator **760** reads the kernel resume event from the power event queue. The power coordinator **760** may then send messages indicating the source of the kernel resume to the multiple client programs via the IPC mechanisms (FIGS. 7A and 7B).

VIII. Conclusion

[0236] The description above discloses, among other things, various example systems, methods, apparatus, and articles of manufacture including, among other components, firmware and/or software executed on hardware. It is understood that such examples are merely illustrative and should not be considered as limiting. For example, it is contemplated that any or all of the firmware, hardware, and/or software aspects or components can be embodied exclusively in hardware, exclusively in software, exclusively in firmware, or in any combination of hardware, software, and/or firmware. Accordingly, the examples provided are not the only way(s) to implement such systems, methods, apparatus, and/or articles of manufacture.

[0237] The specification is presented largely in terms of illustrative environments, systems,

procedures, steps, logic blocks, processing, and other symbolic representations that directly or indirectly resemble the operations of data processing devices coupled to networks. These process descriptions and representations are typically used by those skilled in the art to most effectively convey the substance of their work to others skilled in the art. Numerous specific details are set forth to provide a thorough understanding of the present disclosure. However, it is understood to those skilled in the art that certain embodiments of the present disclosure can be practiced without certain, specific details. In other instances, well known methods, procedures, components, and circuitry have not been described in detail to avoid unnecessarily obscuring aspects of the embodiments. Accordingly, the scope of the present disclosure is defined by the appended claims rather than the foregoing description of embodiments.

[0238] When any of the appended claims are read to cover a purely software and/or firmware implementation, at least one of the elements in at least one example is hereby expressly defined to include a tangible, non-transitory medium such as a memory, DVD, CD, Blu-ray, and so on, storing the software and/or firmware.

[0239] The present technology is illustrated, for example, according to various aspects described below. Various examples of aspects of the present technology are described as numbered examples (1, 2, 3, etc.) for convenience. These are provided as examples and do not limit the present technology. It is noted that any of the dependent examples may be combined in any combination, and placed into a respective independent example. The other examples can be presented in a similar manner.

[0240] Example 1: A method to be performed by a playback device, the method comprising: launching a power coordinator background process, the power coordinator background process having multiple client programs; establishing respective inter-process communication (IPC) mechanisms between the multiple client programs and the power coordinator background process; receiving, via the established IPC mechanisms from the multiple client programs, messages indicating (a) that the respective client program is ready to suspend and (b) a respective resume time; determining, based on the received messages from the multiple client programs, that each client program of the multiple client programs is ready to suspend; based on determining that each client program of the multiple client programs is ready to suspend, (i) sending instructions to the operating system to kernel suspend and (ii) setting a kernel suspend timeout trigger to the earliest resume time among the resume times indicated in the received messages from the multiple client programs; while in the kernel suspend, detecting a particular trigger to kernel resume from among a plurality of triggers to kernel resume, wherein the plurality of triggers to kernel resume comprise the kernel suspend timeout trigger; and in response to the detecting the particular trigger to kernel resume, performing a kernel resume.

[0241] Example 2: The method of Example 1, wherein the playback device is configured to operate in one of multiple power levels, the multiple power levels comprising a first power level and a second power level, the playback device consuming less power in the second power level relative to the first power level, and wherein the method further comprise: receiving, via the established IPC mechanisms from the multiple client programs, messages indicating respective power level requirements of the multiple client programs; determining, based on the received messages from the multiple client programs, a particular power level from among the multiple power levels, the particular power level being the highest power level among the respective power level requirements of the multiple client programs; and sending instructions to the operating system to operate at the particular power level.

[0242] Example 3: The method of Examples 2, wherein operating at the particular power level comprises at least one of (a) decreasing a frequency of at least one processor of the one or more processors or (b) disabling one or more cores of the at least one processor.

[0243] Example 4: The method of any one of Examples 1-3, further comprising: estimating a system resume time based on a usage history of the playback device, and determining that the

estimated system resume time is at least a threshold period of time in advance of the current time, wherein sending instructions to the operating system to kernel suspend comprises sending instructions to the operating system to kernel suspend further based on determining that the estimated system resume time is at least the threshold period of time in advance of the current time.

[0244] Example 5: The method of any one of Examples 1-4, further comprising: based on determining that each client program the multiple client programs is ready to suspend, sending, via the established IPC mechanisms to the multiple client programs, respective messages indicating that kernel suspend is imminent; waiting a pre-determined time period from sending the respective messages indicating that kernel suspend is imminent; and when each client program of the multiple client programs is ready to suspend after the pre-determined time period, sending the instructions to the operating system to kernel suspend.

[0245] Example 6: The method of any one of Examples 1-5, wherein the particular trigger is a power button event, and wherein detecting the particular trigger to enter the first power mode comprises receiving, from the operating system, data indicating a power button event, the method further comprising: sending, via the established IPC mechanisms to the multiple client programs, respective messages indicating the power button event.

[0246] Example 7: The method of any one of Examples 1-6, wherein the particular trigger is a battery voltage level event, and wherein detecting the particular trigger to enter the first power mode comprises receiving, from the operating system, data indicating the battery voltage level event, the method further comprising: sending, via the established IPC mechanisms to the multiple client programs, respective messages indicating the battery voltage level event.

[0247] Example 8: The method of any one of Examples 1-7, wherein the particular trigger is a wake-on-wireless event, and wherein detecting the particular trigger to enter the first power mode comprises receiving, from the operating system, data indicating the wake-on-wireless event, the method further comprising: sending, via the established IPC mechanisms to the multiple client programs, respective messages indicating the wake-on-wireless event.

[0248] Example 9: The method of any one of Examples 1-8, wherein the multiple client programs comprise a network manager program, the method further comprising: determining, via the network manager program, that the IEEE 802.11-compatible network interface is idle; determining a time that the IEEE 802.11-compatible network interface is expected to receive packets; and based on determining that the IEEE 802.11-compatible network interface is idle, sending, via the established IPC mechanism, a message indicating (i) that the network manager program is ready to suspend and (ii) a resume time that is a pre-determined period of time before the determined time that the IEEE 802.11-compatible network interface is expected to receive packets.

[0249] Example 10: The method of any one of Examples 1-9, wherein the multiple client programs comprise an update manager program, the method further comprising: determining, via the update manager program, that no updates are scheduled; and based on determining that no updates are scheduled, sending, via the established IPC mechanism, a message indicating (i) that the update manager program is ready to suspend and (ii) a resume time that is a pre-determined period of time from a current time.

[0250] Example 11: The method of any one of Examples 1-10, wherein the multiple client programs comprise a control program that configures the playback device to play back audio via the one or more speakers and the one or more amplifiers, the method further comprising: during audio playback via the one or more speakers and the one or more amplifiers, foregoing sending a message indicating that the messages indicating the control plane program is ready to suspend.

[0251] Example 12: The method of any one of Examples 1-11, wherein the multiple client programs comprise a particular subset of client programs, the particular subset of client programs comprising a control program that configures the playback device to play back audio via the one or more speakers and the one or more amplifiers, the method further comprising: preventing kernel suspend when the respective IPC mechanisms between the particular subset of client programs and

the power coordinator background process are disconnected.

[0252] Example 13: The method of any one of Examples 1-12, further comprising: receiving, from the operating system, charge state event data indicating a charge state of the battery, the charge state comprising one of: (a) the battery is connected to a charging source or (b) the battery is disconnected from the charging source; sending, via the established IPC mechanisms to the multiple client programs, messages indicating the charge state of the battery; and receiving, via the established IPC mechanisms from a given client program, a message indicating that the given client program is ready to suspend and a given resume time, wherein the given client program determines the given resume time based on the charge state.

[0253] Example 14: A playback device comprising a control interface comprising a power button and transport controls; one or more speakers; one or more amplifiers configured to drive the one or more speakers; a battery; communications interfaces comprising an IEEE 802.11-compatible network interface and an IEEE 802.15-compatible interface; one or more processors; and a housing carrying the one or more speakers, the one or more amplifiers, the battery, the communications interfaces, the one or more processors, and data storage having stored thereon (i) an operating system configured to kernel suspend and resume and (ii) instructions that are executable by the one or more processors to cause the playback device to perform the method of any of Examples 1-13.

[0254] Example 15: A tangible, non-transitory, computer-readable medium having instructions stored thereon that are executable by one or more processors to cause a playback device to perform the method of any one of Examples 1-13.

[0255] Example 16: A portable playback device comprising: a control interface comprising a power button and transport controls; one or more speakers; one or more amplifiers configured to drive the one or more speakers; a battery; communications interfaces comprising an IEEE 802.11-compatible network interface and an IEEE 802.15-compatible interface; a main system-on-chip (SoC) comprising one or more main processor cores, an auxiliary processor core configured to enable the one or more main processor cores upon receiving an interrupt, and a kernel that executes on the one or more main processor cores, wherein a kernel suspend of the kernel disables the one or more main processor cores; a power management microcontroller configured to perform functions comprising: during kernel suspend of the kernel, monitoring, via one or more sensors of the power management microcontroller, the battery for conditions corresponding to respective wake-on-battery triggers, wherein the wake-on-battery triggers comprise (i) a battery fault, (ii) a critical battery level, and (iii) battery charging initiated; detecting that the monitored conditions correspond to a particular wake-on-battery trigger; and in response to detecting that the monitored conditions correspond to particular wake-on-battery trigger, sending, to the auxiliary processor core, an interrupt corresponding to the particular wake-on-battery trigger, wherein the interrupt causes the auxiliary processor core to enable the one or more main processor cores and resume the kernel from kernel suspend; and a housing carrying the one or more speakers, the one or more amplifiers, the battery, the communications interfaces, the power management microcontroller, and the main SoC, wherein the kernel is configured to perform functions comprising: after resuming from kernel suspend, adding a first kernel resume source event to a power event queue, the first kernel resume source event indicating the particular wake-on-battery trigger, wherein a power coordinator background process is configured to read the first kernel resume source event from the power event queue and send data indicating the particular wake-on-battery trigger to one or more client programs via one or more inter-process communication (IPC) mechanisms.

[0256] Example 17: The portable playback device of Example 16: wherein the power management microcontroller is configured to perform functions further comprising: during kernel suspend of the kernel, monitoring, via the one or more sensors of the power management microcontroller, a circuit for a signal corresponding to a wake-on-button trigger, the signal indicating a press of the power button; detecting that the monitored circuit has received the signal corresponding to the wake-on-button trigger; and in response to detecting that the monitored circuit has received the signal

corresponding to the wake-on-button trigger, sending, to the auxiliary processor core, an interrupt corresponding to the wake-on-button trigger, wherein the interrupt causes corresponding to the wake-on-button trigger the auxiliary processor core to enable the one or more main processor cores and resume the kernel from kernel suspend.

[0257] Example 18: The portable playback device of any one of Examples 16-17: wherein the portable playback device further comprises a secondary SoC configured to perform functions comprising: during kernel suspend of the kernel, monitoring, via an 802.15-compatible Bluetooth Low Energy (BLE) interface, for a wake-on-BLE trigger, wherein monitoring for the wake-on-BLE trigger comprises (i) determining whether a BLE connection request has been received and (ii) after receiving the BLE connection request, determining whether a BLE connection was successful; and in response to determining that the BLE connection was successful, sending, to the auxiliary processor core, an interrupt corresponding to the wake-on-BLE trigger, wherein the interrupt corresponding to the wake-on-BLE trigger causes the auxiliary processor core to enable the one or more main processor cores and resume the kernel from kernel suspend.

[0258] Example 19: The portable playback device of any one of Examples 16-18: wherein the power coordinator background process is configured to read a second kernel resume source event from the power event queue, the second kernel resume source event indicating the wake-on-BLE trigger and send data indicating the wake-on-BLE trigger to the one or more client programs via the one or more inter-process communication (IPC) mechanisms, and wherein a first client program changes a playback source of the portable playback device to the BLE connection in response to receiving the data indicating the wake-on-BLE trigger.

[0259] Example 20: The portable playback device of any one of Examples 16-19: wherein the playback device further comprises a capacitive-touch user interface comprising the transport controls, and wherein the secondary SoC configured to perform functions further comprising: during kernel suspend of the kernel, monitoring the capacitive-touch user interface for data indicating a wake-on-touch trigger; determining that the monitored capacitive-touch user interface has received a capacitive touch corresponding to the wake-on-touch trigger; and in response to determining that the monitored capacitive-touch user interface has received a capacitive touch corresponding to the wake-on-touch trigger, sending, to the auxiliary processor core, an interrupt corresponding to the wake-on-touch trigger, wherein the interrupt corresponding to the wake-on-touch trigger causes the auxiliary processor core to enable the one or more main processor cores and resume the kernel from kernel suspend.

[0260] Example 21: The portable playback device of any one of Examples 16-20: wherein the portable playback device further comprises an 802.11-compatible Wi-Fi chipset configured to perform functions comprising: during kernel suspend of the kernel, monitoring, via the 802.11-compatible Wi-Fi chipset, for a wake-on-wireless trigger, wherein monitoring for the wake-on-wireless trigger determining whether a wake-on-wireless magic packet has been received via the 802.11-compatible Wi-Fi chipset; and in response to determining that the wake-on-wireless magic packet has been received via the 802.11-compatible Wi-Fi chipset, sending, to the auxiliary processor core, an interrupt corresponding to the wake-on-wireless trigger, wherein the interrupt corresponding to the wake-on-wireless trigger causes the auxiliary processor core to enable the one or more main processor cores and resume the kernel from kernel suspend.

[0261] Example 22: The portable playback device of any one of Examples 16-21: wherein the portable playback device further comprises a real time clock (RTC) chipset configured to perform operations comprising: before kernel suspend of the kernel, receiving, via the kernel, instructions from the power coordinator background process to set a kernel suspend timeout trigger for a particular time; in response to receiving instructions from the power coordinator background process to set the kernel suspend timeout trigger for the particular time, setting the kernel suspend timeout trigger for the particular time; during kernel suspend of the kernel, monitoring a clock of the RTC chipset for the kernel suspend timeout trigger; determining that the monitored clock of the

RTC chipset indicates the particular time; and in response to determining that the monitored clock of the RTC chipset indicates the particular time, sending, to the auxiliary processor core, an interrupt corresponding to the kernel suspend timeout trigger, wherein the interrupt corresponding to the kernel suspend timeout trigger causes the auxiliary processor core to enable the one or more main processor cores and resume the kernel from kernel suspend.

[0262] Example 23: The portable playback device of any one of Examples 16-22: wherein the particular wake-on-battery trigger is the battery fault, and wherein detecting that the monitored conditions correspond to the particular wake-on-battery trigger comprises: detecting, via the one or more sensors of the power management microcontroller, data indicating that (a) a battery charge level of the battery has decreased below a minimum threshold for operating the portable playback device or (b) a temperature of the battery has exceeded a maximum operating temperature.

[0263] Example 24: The portable playback device of any one of Examples 16-23: wherein the particular wake-on-battery trigger is the critical battery level, and wherein detecting that the monitored conditions correspond to the particular wake-on-battery trigger comprises: detecting, via one or more sensors of the power management microcontroller, that a battery charge level of the battery has decreased below the critical battery level.

[0264] Example 25: The portable playback device of any one of Examples 16-24: wherein the particular wake-on-battery trigger is battery charging initiated, and wherein detecting that the monitored conditions correspond to the particular wake-on-battery trigger comprises: detecting, via one or more sensors of the power management microcontroller, one or more of (a) that a battery charge level of the battery is increasing, (b) the portable playback device has been placed upon a charging base, or (c) that a charging cable has been connected to a port of the portable playback device.

[0265] Example 26: A method to perform the functions of the portable playback device of any one of Examples 16-25.

[0266] Example 27: A tangible, non-transitory, computer-readable medium having instructions stored thereon that are executable by one or more processors to cause a portable playback device to perform the functions of any one of Examples 16-25.

Claims

1. (canceled)

2. A headphone device comprising: one or more speakers; one or more amplifiers configured to drive the one or more speakers; a communications interface configured to facilitate communication over at least one network; one or more processors; and at least one non-transitory computer-readable medium comprising program instructions that are executable by the one or more processors such that the headphone device is configured to execute an operating system, execute one or more programs including a first program that facilitates playback of audio content by the headphone device, suspend execution of the one or more programs and the operating system, detect one or more triggers of a plurality of triggers to resume execution of the one or more suspended programs and the suspended operating system, the plurality of triggers comprising a wake-on-Bluetooth trigger, a wake-on-touch trigger, a timeout trigger, and a power button trigger, resume, based on detection of the one or more triggers, execution of the one or more suspended programs and the suspended operating system, and play back, via execution of the first program, the audio content through the one or more speakers and the one or more amplifiers.

3. The headphone device of claim 2, wherein: the program instructions comprise program instructions executable by the one or more processors to cause the headphone device to determine a current time, and estimate a system resume time based on a usage history of the headphone device; and to suspend execution of the one or more programs and the operating system comprises to suspend execution based on a determination that the system resume time is at least a threshold

period of time in advance of the current time.

4. The headphone device of claim 2, wherein the program instructions comprise program instructions executable by the one or more processors to cause the headphone device to: generate one or more messages via execution of the one or more programs, the one or more messages indicating one or more resume times of the one or more programs; determine, based on the one or more messages, that each program of the one or more programs is ready to suspend execution; and set the timeout trigger based on the one or more resume times.

5. The headphone device of claim 4, wherein: the one or more programs comprise second program that facilitates communication; and the program instructions comprise program instructions executable by the one or more processors to cause the headphone device to determine, via execution of the second program, that the communications interface is idle, determine a time that the communications interface is expected to receive packets, and generate a message of the one or more messages that indicates (i) the second program is ready to suspend, and (ii) a resume time that is a pre-determined period of time before the time that the communications interface is expected to receive packets.

6. The headphone device of claim 4, wherein: the one or more programs comprise an update manager program; and the program instructions comprise program instructions executable by the one or more processors to cause the headphone device to determine, via execution of the update manager program, that no updates are scheduled, and generate a message of the one or more messages that indicates the update manager program is ready to suspend, and a resume time that is a pre-determined period of time from a current time.

7. The headphone device of claim 4, wherein the program instructions comprise program instructions executable by the one or more processors to cause the headphone device to forego generation of at least one message of the one or more messages during execution of the first program.

8. The headphone device of claim 4, wherein: the headphone device is configured to operate at one of a plurality of power levels comprising a first power level and a second power level, the first power level being greater than the second power level; the one or more messages indicate one or more power level requirements of the one or more programs; and the program instructions comprise program instructions executable by the one or more processors to cause the headphone device to determine, based on the one or more messages, a highest power level requirement from the one or more power level requirements, and operate at a power level of the plurality of power levels that supports the highest power level requirement.

9. The headphone device of claim 4, further comprising: a battery; and wherein the plurality of triggers further comprises a battery voltage level trigger.

10. The headphone device of claim 9, wherein: the program instructions comprise program instructions executable by the one or more processors to cause the headphone device to determine a charge state of the battery; and to generate the one or more messages comprises to generate at least one message indicating at least one resume time based on the charge state.

11. The headphone device of claim 2, wherein the headphone device comprises one of: over-the-ear headphones, on-ear headphones, or in-ear earphones.

12. A method of power management method executed by a headphone device, the method comprising: executing an operating system; executing one or more programs comprising a first program that facilitates playback of audio content by the headphone device; suspending execution of the one or more programs and the operating system; detecting one or more triggers of a plurality of triggers to resume execution of the one or more suspended programs and the suspended operating system, the plurality of triggers including a wake-on-Bluetooth trigger, a wake-on-touch trigger, a power button trigger, a battery voltage level trigger, and a timeout trigger; resuming, based on detection of the one or more triggers, execution of the one or more suspended programs and the suspended operating system; and playing back, via execution of the first program, the audio

content via one or more speakers and one or more amplifiers.

13. The method of claim 12, further comprising: determining a current time; and estimating a system resume time based on a usage history of the headphone device, wherein suspending execution comprises suspending execution based on a determination that the system resume time is at least a threshold period of time in advance of the current time.

14. The method of claim 12, further comprising generating at least one message indicating imminent suspension of execution of the one or more programs and the operating system, wherein suspending execution comprises suspending execution after expiration of at least a pre-determined time period from communication of the at least one message.

15. The method of claim 12, further comprising: generating one or more messages via execution of the one or more programs, the one or more messages indicating one or more resume times of the one or more programs; determining, based on the one or more messages, that each program of the one or more programs is ready to suspend execution; and setting the timeout trigger based on an earliest time of the one or more resume times.

16. The method of claim 15, wherein the one or more programs comprise a second program, the method further comprising: determining, via execution of the second program, that a communications interface of the headphone device is idle; and determining a time that the communications interface is expected to receive packets; wherein generating the one or more messages includes generating a message that indicates that the second program is ready to suspend; and wherein the one or more resume times includes a network resume time that is a pre-determined period of time before the time that the communications interface is expected to receive packets.

17. The method of claim 15, wherein: generating the one or more messages comprises generating one or more messages indicating one or more power level requirements of the one or more programs; and the method comprises determining based on the one or more messages, a highest power level requirement from the one or more power level requirements, and operating the headphone device at a power level of a plurality of power levels that supports the highest power level requirement.

18. A non-transitory computer readable medium storing instructions executable by one or more processors to cause a headphone device to execute a power management method, the instructions comprising instructions to: execute an operating system; execute one or more programs, at least one program of the one or more programs being a first program that facilitates playback of audio content by the headphone device; suspend execution of the one or more programs and the operating system; detect one or more triggers of a plurality of triggers to resume execution of the one or more suspended programs and the suspended operating system, the plurality of triggers including a wake-on-Bluetooth trigger, a wake-on-touch trigger, a power button trigger, a battery voltage level trigger, and a timeout trigger; resume, based on detection of the one or more triggers, execution of the one or more suspended programs and the suspended operating system; and playback, via execution of the first program, the audio content through one or more speakers and one or more amplifiers of the headphone device.

19. The non-transitory computer readable medium of claim 18, wherein the one or more programs comprise a second program, and wherein the instructions further comprise instructions to: determine, via execution of the second program, that a communications interface of the headphone device is idle; determine a time that the communications interface is expected to receive packets; and generate a message that indicates (i) the second program is ready to suspend, and (ii) a resume time that is a pre-determined period of time before the time that the communications interface is expected to receive packets.

20. The non-transitory computer readable medium of claim 18, wherein the instructions further comprise instructions to: generate one or more messages via execution of the one or more programs, the one or more messages indicating one or more resume times of the one or more programs; determine, based on the one or more messages, that each program of the one or more

programs is ready to suspend execution; and set the timeout trigger based on the one or more resume times.
